

The Linear Regression Model

Lecture 1

The plan of this lecture is to offer a review of regression analysis for the conditional mean function $E(y|x_j, x_{-j})$. We will introduce the basic regression model and then concentrate in the simple case of a bivariate regression. At the end of the lecture, we discuss statistical properties of the Ordinary Least Squares (OLS) estimator.

1 Multivariate Regression Analysis

We start assuming a linear model with p independent variables,

$$\underbrace{y}_{1 \times 1} = \underbrace{x'}_{1 \times p} \underbrace{\beta}_{p \times 1} + \underbrace{u}_{1 \times 1} \quad (1)$$

where y is the response or dependent variable and the β 's are the coefficients associated with the independent variables x 's. At the core of econometrics, there is the concept of causal effects or causal relationships. Obviously, we are not referring to the notion of correlation between x_j and y since $\text{Corr}(x_j, y) = \text{Corr}(y, x_j)$.

We define x_j to be an independent variable, y to be a dependent variable and we might have other (independent) variables we keep at a constant level, x_{-j} . To investigate causal relationships we employ models and the classical model of interest is the conditional expected value of the dependent variable conditional on the independent variables,

$$E(y|x_j, x_{-j}) = E(y|x) = \int y f_{Y|X=x} dx.$$

We can also define:

$$\frac{\partial E(y|x_j, x_{-j})}{\partial x_j} := \text{partial effect}$$

obtained by keeping x_{-j} equal to a constant. Note that this is well defined if x_j is a continuous variable.

To estimate causal effects, a crucial assumption is that all observable factors (the independent variables $x_1, x_2, \dots, x_p$) that affect the dependent variable y are uncorrelated with

the error term.

$$E(u|x_1, x_2, \dots, x_p) = 0. \quad (2)$$

If condition (2) is satisfied, we say that the variables $\{x_1, x_2, \dots, x_p\}$ are strictly exogenous. As explained below, the implication is that the error term has mean equal to zero.

$$E(u|x) = E(u) = 0.$$

To summarize, the first two assumptions of the classical regression model are:

A1 (linearity) The model is linear as in (1)

A2 (strict exogeneity) We assume that $E(u|x_1, x_2, \dots, x_p) = 0$

We will use later a model with p independent variables as in equation (1). Let us now concentrate in one particular case assuming that we have one independent variable, $p = 1$.

2 Conditional Expectations

The simple regression model is,

$$y = \beta_0 + \beta_1 x + u \quad (3)$$

where y is the dependent variable, x is the independent variable, and u is the error term or disturbance. The model suggests that changes in y are induced by changes in x and u . We want to know how much y can be explained in terms of x .

Example 1. Suppose we want to know the effect of education on wages and wages and education are linearly related. Then, we can write

$$wages = \beta_0 + \beta_1 educ + u \quad (4)$$

In this example, we assume that the relation between the response variable and the independent variable is *linear in the parameters* β_0 and β_1 . It may be the case that the model is

$$\log(wages) = \beta_0 + \beta_1 educ + \beta_2 educ^2 + u,$$

which is *non linear in variables*.

We can define the population regression function (PRF) or, in this case, the population regression Line (PRL) of equation (3) as,

$$E(y|x) = E(\beta_0 + \beta_1 x + u|x) = \beta_0 + \beta_1 x + E(u|x) = \beta_0 + \beta_1 x, \quad (5)$$

by Assumptions A1 and A2. It says that the expected value of the dependent variable for a given value of the independent variables is equal to the parameter β_0 plus β_1 times the independent variable. In the case of the PRL with one slope parameter, the model contains two unknown parameters β_0 (the intercept) and β_1 (the slope).

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Our interest lies on estimating the PRF, which implies estimating the two unknowns, i.e. the intercept and the slope.

We now discuss the implications of the previous conditions A1 and A2. We can write the model as:

$$y = E(y|x) + u \quad (6)$$

$$E(u|x) = 0. \quad (7)$$

Equation (7) implies:

$$E(u) = 0 \quad (8)$$

$$Cov(u, x_j) = 0, \text{ for } j = 1, \dots, p, \quad (9)$$

$$Cov(u, h(x_j)) = 0, \text{ for } j = 1, \dots, p, \quad (10)$$

where Cov denotes covariance and $h(\cdot)$ is a known function (e.g. $h(x_1) = x_1^2$, $h(x_1, x_2) = x_1 x_2^2$, $h(x_2) = \exp(x_2 + x_2^2)$, etc.). To see these results, first note that by definition $u := y - E(y|x)$. Then,

$$E(u) = E(y - E(y|x)) = E(y) - E(E(y|x)) = E(y) - E(y) = 0. \quad (11)$$

To show that also implies independence between the error term and covariates, we need to show that

$$E(ux) = E(E(ux|x)) = E(xE(u|x)) = E(x \times 0) = 0. \quad (12)$$

Therefore, conditions (11) and (12) imply, for $j = 1, \dots, p$,

$$Cov(u, x_j) = E(xu) - E(x)E(u) = 0. \quad (13)$$

3 Best Linear Predictor

Define a loss function of a generic form, $L(u) = L(y - m(x))$, where $m(x)$ is the predictor in terms of x and u is the prediction error. Naturally, a decision maker tries to solve the following problem:

$$\min_{m \in \mathcal{M}} E\{L(y - m(x))|x\},$$

that is, she would like to choose a predictor of y that minimizes the expected loss.

It is convenient to assume a quadratic loss (e.g. $L(u) = u^2$). Therefore the problem may be transformed to minimize the mean square error of $m(x)$ for y ,

$$\begin{aligned} E\{(y - m(x))^2|x\} &= E\{(y - \mu(x) + \mu(x) - m(x))^2|x\} \\ &= V(y|x) + (\mu(x) - m(x))^2 + 2E[(m(x) - \mu(x))(y - \mu(x))|x] \\ &= V(y|x) + (\mu(x) - m(x))^2 \end{aligned}$$

where $V(y|x)$ is the conditional variance of y .

If the first two conditional moments of y exists (e.g., $E(y|x) < \infty$ and $V(y|x) < \infty$), the (unconditional) mean square error is minimized to the conditional variance of y when $m(x)$ is equal to the conditional mean $\mu := E(y|x)$,

$$E(y - m(x))^2 = EV(y|x) + E(m(x) - \mu(x))^2.$$

Suppose now that the functional form of m is known. For instance, consider now that a model linear in parameters:

$$\min_{\beta} E\{(y - x'\beta)^2\} = \min_{\beta} \{Ey^2 - 2E(yx')\beta + \beta'E(xx')\beta\}$$

Taking first order condition, we have

$$-E(xy) + E(xx')\beta = 0,$$

thus,

$$\beta = (E(xx'))^{-1}E(xy)$$

Therefore, the parameter β can be interpreted as the best linear predictor under squared error loss.

Note that for β to exists, we need the $p \times p$ matrix $E(xx')$ to be invertible. (We also need that $E(y^2) < \infty$ and $E(\sum_j x_j^2) < \infty$ (see Hansen 2017, Theorem 2.18.1)). If no redundant variables are included in the vector x , then the design matrix $E(xx')$ is invertible and we can say that β is unique and *identified*. A requirement is that $E(xx')$ is positive definite. Identification in this class of linear models means that we can write the parameter of interest in terms of moment conditions that are functions of observed variables such us y and $x_1, \dots, x_p$.

This is related to another assumption of the linear model. The condition

A3 (full rank design matrix) The rank $(E(x_i x_i')) = p$

This condition rules out perfect collinearity, or in other words, it rules out exact linear

relationships among the independent variables.

4 Partial Effects and Elasticities

Consider that we have the following population regression model,

$$E(y|x) = \beta_0 + \sum_{j=1}^p \beta_j x_j,$$

and we are interested in the marginal effect of the j -th independent variable on the conditional expected value of y ,

$$\frac{\partial E(y|x)}{\partial x_j} = \beta_j.$$

This tells us the increase in the mean value of y when the independent variable is increased by 1 unit, other things equal. This interpretation is equivalent to the notion of *ceteris paribus*: the effect of x on y holding everything constant. Note that if x_i and x_j are related, finding the marginal effect is somewhat more complicated, but the linearity of the regression model has a simple and sometimes useful implication for empirical analysis.

Consider the following model with $p = 3$ and $x_3 = x_1 \times x_2$:

$$E(y|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 \times x_2).$$

where $\mathbf{x} = (x_1, x_2, x_3)'$. The partial effect of x_1 on the conditional expected value of the dependent variable is:

$$\frac{\partial E(y|\mathbf{x})}{\partial x_1} = \beta_1 + \beta_3 x_2,$$

and,

$$\frac{\partial E(y|\mathbf{x})}{\partial x_2} = \beta_2 + \beta_3 x_1.$$

Sometimes we are interested in the elasticity of y with respect to x_j :

$$\frac{\partial y}{\partial x_j} \frac{x_j}{y}$$

Our partial elasticity is,

$$\frac{\partial E(y|\mathbf{x})}{\partial x_j} \frac{x_j}{E(y|\mathbf{x})} = \frac{\partial \log(E(y|\mathbf{x}))}{\partial \log(x_j)}$$

For instance, the partial elasticity with respect to x_1 in our previous example is:

$$\frac{(\beta_2 + \beta_3 x_2)x_1}{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 \times x_2}$$

When u and x are not correlated, as it is the case in the previous models, we can have an alternative definition of elasticity that it is more convenient:

$$\frac{\partial E(\log(y)|\mathbf{x})}{\partial \log(x_j)}$$

Note that $E(y|x) = \exp(\mathbf{x}'\boldsymbol{\beta})$ implies,

$$\log(E(y|\mathbf{x})) = \mathbf{x}'\boldsymbol{\beta}.$$

If $y = \exp(\mathbf{x}'\boldsymbol{\beta} + u)$, then $\log(y) = \mathbf{x}'\boldsymbol{\beta} + u$, and by previous assumptions $E(\log(y)) = \mathbf{x}'\boldsymbol{\beta}$ because $E(u|\mathbf{x}) = 0$.

Therefore, the parameter β_j should be interpreted as an elasticity if the model is in a ‘log-log’ form,

$$\log(y) = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + u$$

or as a semielasticity if the model is a log-linear model:

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

which implies that the partial effect is

$$\frac{\partial E(\log(y)|\mathbf{x})}{\partial x_j}.$$

Lastly, we can introduce the concept of average partial effects (for details, see Blundell and Powell (2003)). Suppose we have the following model,

$$E(y|\mathbf{x}) = g(\mathbf{x}, z),$$

where $\mathbf{x}$ is an observed variable and z is a latent variable. The partial effect is:

$$\frac{\partial E(y|\mathbf{x}, z)}{\partial x_j} = \frac{\partial g(\mathbf{x}, z)}{\partial x_j}$$

Note that the partial effect depends on a latent variable which can take different values from

an underlying distribution. The average partial effect at $\mathbf{x} = \mathbf{x}_0$ is,

$$\delta_j(\mathbf{x}_0) = E\left(\frac{\partial g(\mathbf{x}_0, z)}{\partial x_j}\right) = \int \frac{\partial g(\mathbf{x}_0, z)}{\partial x_j} f(z) dz$$

Can we estimate the average partial effect if we do not observe z ? We essentially have two important cases. If $\mathbf{x}$ and z are independent, the average partial effect is simply the partial effect with respect to x_j :

$$\frac{\partial E(y|\mathbf{x})}{\partial x_j}.$$

Alternatively, if we want to model unobserved heterogeneity, we can find $E(y|\mathbf{x})$ by integrating out over the distribution of the latent variable z .

Example 2. Suppose,

$$y = \beta_0 + \beta_1 x_1 + \gamma_1 x_1 z_1 + u$$

where $E(u|x_1, z_1) = 0$. The conditional mean model is

$$E(y|x, z) = \beta_0 + \beta_1 x_1 + \gamma_1 x_1 z_1$$

and,

$$\frac{\partial E(y|x, z)}{\partial x_j} = \beta_1 + \gamma_1 z_1.$$

If $E(z_1) = E(z_1|x_1) = 0$, then the average partial effect is,

$$E(\beta_1 + \gamma_1 z_1) = \beta_1 = \frac{\partial E(y|x)}{\partial x_j}.$$

If $\mathbf{x}$ and z are not independent, but they are conditional independent on a proxy variable $\mathbf{w}$, it is possible to show that the average partial effect is:

$$\delta_j(\mathbf{x}_0) = E_z\left(\frac{\partial E(y|\mathbf{x}_0, z)}{\partial x_j}\right) = E_z\left(\frac{\partial E(y|\mathbf{x}_0, \mathbf{w})}{\partial x_j}\right).$$

5 Estimation

Consider a random sample $\{y_i, x_i\}$ of size n from population. The pair of data points (y_i, x_i) represent the i^{th} observation. The size of the sample n denotes the total number of observations. Since each data point comes from model (1),

$$y_i = \beta_0 + \beta_1 x_i + u_i \tag{14}$$

where $i = 1, \dots, n$. Our goal is to estimate the parameters (β_0, β_1) .

5.1 Method of Moments

We first review an estimation method called Method of Moments using the following example.

Example 3. Suppose you want to estimate the first moment of a random variable. The method of moments (MM) can be used to estimate the population mean. Consider $\{y_i\}$ to be an independently and identically distributed (i.i.d.) random variable drawn from a distribution whose first moment exists (e.g., y is not distributed as Cauchy). In the population,

$$E(y - \mu) = 0$$

so we can use the corresponding sample moment condition

$$\frac{1}{n} \sum_{i=1}^n (y_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{y}.$$

To obtain the MM estimator for the linear model, recall the moment condition,

$$E(x_i u_i) = E(E(x_i u_i | x)) = E(x E(u_i | x)) = E(x \times 0) = 0,$$

or alternatively,

$$E(x_i(y_i - x_i' \beta)) = 0,$$

where $x_i = (1, x_{i1})'$ and $\beta = (\beta_0, \beta_1)'$. The MM estimator could be obtained as the solution of the corresponding sample moment condition,

$$\frac{1}{n} \sum_{i=1}^n (x_i(y_i - x_i' \beta)) = 0,$$

yielding,

$$\hat{\beta} = \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum_{i=1}^n x_i y_i.$$

We now turn to some basic linear regression models.

5.2 Ordinary Least Squares Estimation (OLS):

Our model's assumptions were

$$\begin{aligned} E(u) &= E(y - \beta_0 - \beta_1 x) = 0 \\ \text{Cov}(x, u) &= E(xu) - E(x)E(u) = E(xu) = E(x(y - \beta_0 - \beta_1 x)) = 0 \end{aligned}$$

These two equations are restrictions imposed to the data about the joint probability distribution of x and y . There are two unknowns and two equations, which suggests that we can solve for β_0 and β_1 . The sample counterpart of these equations are

$$\frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (15)$$

$$\frac{1}{n} \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (16)$$

The Ordinary Least Squares Estimator is the argument that minimizes the following criterion function:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2, \quad (17)$$

with respect to the unknowns β_0 and β_1 . It is easy to see that the normal equations corresponding to this problem are identical to the two previous equations (15) and (16). Thus, the OLS estimator is a MM estimator in this case.

From equation (15), we obtain

$$\bar{y} - \beta_0 - \beta_1 \bar{x} = 0 \Rightarrow \hat{\beta}_0 = \bar{y} - \beta_1 \bar{x} \quad (18)$$

where $\bar{y} = n^{-1} \sum_{i=1}^n y_i$. Thus, if we know the ordinary least square estimator of the slope β_1 , we can obtain an estimate of the intercept. How can we obtain $\hat{\beta}_1$? Plug the last result in equation (16) to obtain,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n x_i (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \beta_1 x_i) &= 0 \\ \frac{1}{n} \sum_{i=1}^n x_i ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})) &= 0 \\ \sum_{i=1}^n x_i (y_i - \bar{y}) &= \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x}) \end{aligned}$$

It can be shown that the last equation is equal to

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})^2$$

which implies that the estimator for the parameter β_1 is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (19)$$

Therefore, $\hat{\beta}_0, \hat{\beta}_1$ are the ordinary least squares (OLS) estimators for the intercept and slope parameters of the simple linear regression model.

5.3 Arithmetic Properties of OLS

It is useful to briefly discuss properties of the OLS estimators. Consider the following 4 properties:

1. The sum of the residuals $\hat{u}_i$ is zero since $\hat{\beta}_0$ is chosen such that

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = \sum_{i=1}^n \hat{u}_i = 0$$

2. The sample covariance between the independent variable and the residuals are zero.
To see this, it is crucial to find that

$$\begin{aligned} \sum_{i=1}^n x_i \hat{u}_i &= \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = \sum_{i=1}^n x_i (y_i - x_i' \hat{\beta}) \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i x_i' \hat{\beta} \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i x_i' \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum_{i=1}^n x_i y_i \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i y_i = 0. \end{aligned}$$

3. The sample average of the fitted values is equal to sample average of the observations.

$$\begin{aligned} y_i &= \hat{y}_i + \hat{u}_i \\ \sum y_i &= \sum \hat{y}_i + \sum \hat{u}_i \\ \frac{1}{n} \sum y_i &= \frac{1}{n} \sum \hat{y}_i \Rightarrow \bar{y} = \bar{\hat{y}} \end{aligned}$$

4. The point $(\bar{x}, \bar{y})$ is always on the regression OLS line.

$$\bar{y} = \bar{\hat{y}} = \frac{1}{n} \sum \hat{y}_i = \frac{1}{n} \sum (\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

5.4 Bias of the OLS estimator

Under condition A1 and A2, the OLS estimator is unbiased, that is, the expected value of the OLS estimator is equal to the unknown parameter.

We can show the result as follows. Recall that,

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) y_i \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) (\beta_0 + \beta_1 x_i + u_i) \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \left(\beta_0 \sum_{i=1}^n (x_i - \bar{x}) + \beta_1 \sum_{i=1}^n (x_i - \bar{x}) x_i + \sum_{i=1}^n (x_i - \bar{x}) u_i \right) \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \left(\beta_1 \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n (x_i - \bar{x}) u_i \right). \end{aligned}$$

Therefore, the OLS estimator for the slope can be written as:

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) u_i. \quad (20)$$

Taking expectation,

$$E(\hat{\beta}_1) = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n E((x_i - \bar{x}) u_i)$$

Since $E(u|x) = E(u) = 0$, $\Rightarrow E(\hat{\beta}_1) = \beta_1$. Also, recall that

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} = \beta_0 + \beta_1 \bar{x} - \hat{\beta}_1 \bar{x} \\ \Rightarrow E(\hat{\beta}_0) &= \beta_0 + \beta_1 \bar{x} - E(\hat{\beta}_1) \bar{x} = \beta_0\end{aligned}$$

Therefore, the OLS estimators are unbiased e.g.

$$E(\hat{\beta}_0) = \beta_0; \quad E(\hat{\beta}_1) = \beta_1 \quad (21)$$

5.5 Variance of the OLS estimator

We are going to obtain the variance of the OLS estimators, but we have to state first one of the assumptions of the simple linear regression model:

A4 (homocedasticity) $\text{Var}(u|x) = \sigma^2$

A5 (uncorrelated errors) $\text{Cov}(u_i, u_j) = 0$ for $i, j = 1, \dots, i-1, i+1, \dots, n$

In other words, the variance of the error term is constant (it does not change with x) and the error u_i is independent of u_j for $j \neq i$. Assumptions A4 and A5 together imply spherical errors, or more precisely, spherical error variance.

What is the variance of $\hat{\beta}_1$? We know that

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) u_i$$

Considering A4 and A5, we obtain the variance as follows:

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var} \left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) u_i \right) \\ &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \text{Var} \left(\sum_{i=1}^n (x_i - \bar{x}) u_i \right)\end{aligned}$$

$$\begin{aligned}
\text{Var}(\hat{\beta}_1) &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \sum_{i=1}^n \text{Var}((x_i - \bar{x})u_i) \\
&= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(u_i) \\
\Rightarrow \text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)}
\end{aligned}$$

Similary, to obtain the variance of the intercept, under the stated assumptions, we write,

$$\begin{aligned}
\text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) = \text{Var}\left(\bar{y} - \frac{\sum_i (x_i - \bar{x}) y_i}{\sum_i (x_i - \bar{x})^2} \bar{x}\right) \\
&= \text{Var}\left(\sum_i \left(\frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right) y_i\right) \\
&= \sum_i \left(\frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right)^2 \text{Var}(u_i) \\
&= \sigma^2 \sum_i \left(\frac{1}{n^2} + \left(\frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right)^2 - \frac{2}{n} \frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right) \\
&= \sigma^2 \sum_i \left(\frac{1}{n^2} + \left(\frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right)^2\right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2}\right) = \sigma^2 \left(\frac{\sum_i (x_i - \bar{x})^2 + n\bar{x}^2}{n \sum_i (x_i - \bar{x})^2}\right) \\
&= \sigma^2 \left(\frac{\sum_i x_i^2 + n\bar{x}^2 - 2\bar{x} \sum_i x_i + n\bar{x}^2}{n \sum_i (x_i - \bar{x})^2}\right) \\
&= \sigma^2 \left(\frac{\sum_i x_i^2 + n\bar{x}^2 - 2n\bar{x}^2 + n\bar{x}^2}{n \sum_i (x_i - \bar{x})^2}\right) = \sigma^2 \left(\frac{\sum_i x_i^2}{n \sum_i (x_i - \bar{x})^2}\right),
\end{aligned}$$

which leads to the variance of the estimated intercept,

$$\text{Var}(\hat{\beta}_0) = \frac{n^{-1} \sum_{i=1}^n x_i^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)} \sigma^2$$

Although we do not know the parameter σ^2 , we can plugin its estimator $\hat{\sigma}^2$.

Remark 1. The variance of the error term is another unknown parameter of the population, but we can estimate using the analog principle. Recall that u is distributed as a normal

variable with mean zero and variance σ^2 ,

$$E(u) = 0; \quad V(u) = E(u^2) - (E(u))^2 = E(u^2) = \sigma^2,$$

which implies that we can use (from the sample counterpart),

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2$$

as an estimator of σ^2 . Unfortunately, this is a biased estimator, but we can easily correct this to have an unbiased estimator of the variance of the error term,

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2 \quad (22)$$

where $(n-2)$ are the degrees of freedom.

Therefore, the estimator of the variance of the OLS estimator,

$$\text{Var}(\hat{\beta}_0) = \frac{n^{-1} \sum_{i=1}^n x_i^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)} \hat{\sigma}^2; \quad \text{Var}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)} \quad (23)$$

Then, we can define the standard error of the slope coefficient as

$$\text{se}(\hat{\beta}_1) = \frac{1}{\sqrt{(\sum_{i=1}^n (x_i - \bar{x})^2)}} \hat{\sigma} \quad (24)$$

6 Multivariate Linear Models

So far, we studied the a simple regression model, which includes only one covariate. It makes sense to extend the previous model to a model with p independent variables,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + u = \beta_0 + \sum_{j=1}^p \beta_j x_j + u \quad (25)$$

where β_0 is the intercept, β_1 is the coefficient associated with the independent variable x_1 , β_2 with the variable x_2 , and β_p with the variable x_p . The random variable u is the error term, which contains other factors than $x_1, x_2, \dots, x_p$.

The crucial assumption is that all observable factors (the independent variables $x_1, x_2, \dots, x_p$) that affect the dependent variable y are uncorrelated with the error term: $E(u|x_1, x_2, \dots, x_p) = 0$.

6.1 Method of Moments

The basic idea here is to apply the restrictions in the population to the data. For instance, recall that we have the assumptions

$$E(u_i) = 0$$

$$E(x_{i1}, u_i) = 0$$

$$E(x_{ip}, u_i) = 0$$

Therefore, we should find the coefficient estimates that solve the following system of equations,

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{i1} \hat{u}_i = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{ip} \hat{u}_i = 0$$

Again, we have $(p+1)$ equations and $(p+1)$ unknowns. The solutions are called the method of moment estimators for the regression parameters $\beta_0, \dots, \beta_p$.

For instance, consider the model with two independent variables. The conditions will be

$$E(u_i) = 0$$

$$E(x_{i1}u_i) = 0$$

$$E(x_{i2}u_i) = 0$$

consequently,

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}) = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{i1} \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n x_{i1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}) = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{i2} \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n x_{i2} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}) = 0$$

As we will see below, the method of moments gives estimates coefficients that are identical to the least squares estimates.

6.2 OLS

The least squares procedure is similar than the one discussed before. It computes the estimated coefficients as minimizers of the sum of residuals squared,

$$S = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The difference is now that

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + u_i$$

instead of the sample $y_i = \beta_0 + \beta_1 x_i$. For instance, the predicted value of y_i is,

$$\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_{ij},$$

under the standard assumptions. To obtain the solution of this problem, we use partial derivatives to develop a set of $(p+1)$ simultaneous normal equations,

$$\begin{aligned} \frac{\partial S}{\partial \hat{\beta}_0} &= 0 \\ \frac{\partial S}{\partial \hat{\beta}_1} &= 0 \\ &\vdots \\ \frac{\partial S}{\partial \hat{\beta}_p} &= 0 \end{aligned}$$

which has $(p+1)$ unknowns $\hat{\beta}_0, \dots, \hat{\beta}_p$. For simplicity, we focus on the problem of minimizing

$$S = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2,$$

with partial derivatives

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n x_{i1} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial S}{\partial \beta_2} = -2 \sum_{i=1}^n x_{i2}(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

we can write these equations as

$$\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_{i1} + \hat{\beta}_2 \sum_{i=1}^n x_{i2} &= \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_{i1} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 + \hat{\beta}_2 \sum_{i=1}^n x_{i1}x_{i2} &= \sum_{i=1}^n x_{i1}y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_{i2} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}x_{i2} + \hat{\beta}_2 \sum_{i=1}^n x_{i2}^2 &= \sum_{i=1}^n x_{i2}y_i \end{aligned}$$

It can be shown that the estimate for the intercept coefficient β_0 is,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2.$$

It is somewhat more tedious to find that

$$\begin{aligned} \hat{\beta}_1 &= \frac{S_y(r_{x_1,y} - r_{x_1x_2}r_{x_2y})}{S_{x_1}(1 - r_{x_1x_2}^2)} \\ \hat{\beta}_2 &= \frac{S_y(r_{x_2,y} - r_{x_1x_2}r_{x_1y})}{S_{x_2}(1 - r_{x_1x_2}^2)} \end{aligned}$$

Where r_{x_jy} is the sample correlation between x_j and y , $r_{x_1x_2}$ is the sample correlation between x_1 and x_2 , S_{x_j} is the sample standard deviation for x_j , and S_y is the sample standard deviation for y .

What we can learn from these estimators is that in the multiple regression model the estimate of the regression coefficient β_1 depends on the correlation between x_1 and x_2 , and r_{x_2} and y . If there is no correlation between x_1 and x_2 , (e.g., $r_{x_1,x_2} = 0$), the estimate for the coefficient β is,

$$\begin{aligned} \hat{\beta}_1 &= \frac{S_y r_{x_1y}}{S_{x_1}} \\ &= \frac{\sum_{i=1}^n (y_i - \bar{y})(x_{i1} - \bar{x})}{\sum_{i=1}^n (x_{i1} - \bar{x})^2} \end{aligned}$$

If $r_{x_1,x_2} = 1$, the estimate of the coefficient β_1 will not be defined. We could obtain the same conclusions on $\hat{\beta}_2$. In empirical analysis, it is always important to explore the correlation between the independent variables.

7 Analysis-of-variance and R^2

Consider,

$$\begin{aligned} y_i &= \hat{y}_i + \hat{u}_i \\ y_i - \bar{y} &= (\hat{y}_i - \bar{y}) + \hat{u}_i \\ \sum (y_i - \bar{y})^2 &= \sum (\hat{y}_i - \bar{y})^2 + 2 \sum \hat{u}_i (\hat{y}_i - \bar{y}) + \sum \hat{u}_i^2 \end{aligned}$$

Note that using the arithmetic properties of the OLS estimators,

$$\sum \hat{u}_i (\hat{y}_i - \bar{y}) = \sum \hat{u}_i (\hat{\beta}_1 (x_i - \bar{x})) = \hat{\beta}_1 (\sum \hat{u}_i x_i - \bar{x} \sum \hat{u}_i) = 0$$

Therefore,

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum \hat{u}_i^2. \quad (26)$$

If we divide the expression by the right hand side variable,

$$1 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} + \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2}$$

which indicates a proportion that it is explained by the linear estimated model (first term) and that it is not explained by the linear model (second term). The R^2 of a regression is defined as

$$R^2 := \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2}$$

Note that a small sum of residuals squared means that the model explains very well, and sum of residuals squared large means that the model does not produce a good fit to the data. Therefore, R^2 close to 1 is telling us that the model produces a good fit to the data, and a R^2 close to 0 says that the model does not produce a good fit to the data. Note, also, that $0 \leq R^2 \leq 1$ since $0 < \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2} < 1$.

An alternative statistic is the adjusted R^2 , which is defined as

$$\bar{R}^2 := 1 - \frac{n-1}{n-p} \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2},$$

and provides a trade-off between goodness-of-fit and parsimony. The goodness-of-fit is measured by the R^2 and the parsimony of the model is measured by the degrees of freedom, $n - p$. Note that the $\bar{R}^2$ may not be well defined and can be negative too.

It is immediately apparent that

$$1 - \bar{R}^2 := \frac{n-1}{n-p} (1 - R^2) .$$

Orthogonal Projections

Lecture 2

We start assuming a linear model,

$$\underbrace{y_i}_{1 \times 1} = \underbrace{x'_i}_{1 \times p} \underbrace{\beta}_{p \times 1} + \underbrace{u_i}_{1 \times 1} \quad \text{for } i = 1, 2, \dots, n, \quad (1)$$

where y is the response variable for subject i and x_i is a vector of p independent variables that includes an intercept. The model can be written in a more convenient matrix notation as

$$\underbrace{y}_{n \times 1} = \underbrace{\underbrace{X}_{n \times p} \underbrace{\beta}_{p \times 1}}_{n \times 1} + \underbrace{u}_{n \times 1} \quad (2)$$

where y is now defined as a $n \times 1$ vector, the matrix X is $n \times p$, the vector of parameters β is $p \times 1$, and the random vector u is $n \times 1$. In other words,

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p-1} \\ 1 & x_{21} & x_{22} & \dots & x_{2p-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np-1} \end{bmatrix}; \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}; \quad u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}.$$

It follows that

$$\sum_{i=1}^n x_i x'_i = X'X,$$

$$\sum_{i=1}^n x_i y_i = X'y.$$

1 Estimation

We can estimate the parameter β using the Method of Moments or alternatively the traditional method called Ordinary Least Squares (OLS). The derivation of the OLS estimator is

based on minimizing the sum of squared errors

$$||u||^2 = ||y - X\beta||^2$$

where $||\cdot||$ is the Euclidean norm defined as $(z'z)^{1/2} = (\sum z_i^2)^{1/2}$ for a generic vector z . Therefore, the objective function is

$$\begin{aligned} \min_{\beta} ||y - X\beta||^2 &= \min_{\beta} (y - X\beta)'(y - X\beta) = \min_{\beta} \{(y' - \beta'X')(y - X\beta)\} \\ &= \min_{\beta} \{y'y - \beta'X'y - y'X\beta + \beta'X'X\beta\} \\ &= \min_{\beta} \{y'y - 2\beta'X'y + \beta'X'X\beta\} \end{aligned}$$

because $\beta'X'y = y'X\beta$. The first order condition is

$$-2X'y + 2X'X\beta = 0,$$

which implies that $X'(y - X\beta) = 0$, and consequently, under Assumptions A1, A2 and A3:

$$\hat{\beta} = (X'X)^{-1}X'y. \quad (3)$$

Alternatively, we define the estimator as

$$\hat{\beta} = \arg \min ||y - X\beta||^2 \quad (4)$$

The OLS estimator $\hat{\beta}$ gives the closest vector $X\hat{\beta}$, in terms of the Euclidean norm, to y among all vectors that lie on the space spanned by the vector X .

INSERT FIGURE

We can also obtain the variance of the estimator under homocedasticity and independent errors (Assumptions A4 and A5). Note that,

$$\begin{aligned} V(\hat{\beta}|X) &= V((X'X)^{-1}X'y|X) = V((X'X)^{-1}X'(X\beta + u)|X) \\ &= (X'X)^{-1}X'V(u|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\sigma_u^2 I_N X(X'X)^{-1} \\ &= \sigma_u^2 (X'X)^{-1}, \end{aligned}$$

under the assumption that the variance of the error term is “spherical” i.e. $V(u|X) = \sigma_u^2 I$. The variance depends on σ_u^2 , which is unknown and needs to be estimated. As in Lecture 1, it is estimated using $\hat{u}'\hat{u}/n - p$.

Note also that the homocedasticity assumption simplifies the expression of the variance

of OLS.

2 Finite Sample Properties

The OLS estimator $\hat{\beta}$ is unbiased when the independent variables are independent of the error term (Assumption A2),

$$\begin{aligned} E(\hat{\beta}|X) &= E((X'X)^{-1}X'y|X) \\ &= \beta + E((X'X)^{-1}X'u|X) \\ &= \beta + (X'X)^{-1}X'E(u|X) \\ &= \beta \end{aligned}$$

By the law of iterated expectations, we also have that

$$E(\hat{\beta}) = E(E(\hat{\beta}|X)) = \beta$$

The OLS estimator is unbiased and linear, which means that we can write $\hat{\beta}$ as a linear function of the response variable y e.g.

$$\hat{\beta} = Ay$$

where $A = (X'X)^{-1}X'$. Now, consider another unbiased linear estimator,

$$\tilde{\beta} = (A + B)y$$

Therefore, it must be the case that $E\tilde{\beta} = \beta$ because $BX = 0$. Also, the variance of the estimator $\tilde{\beta}$ is,

$$\begin{aligned} V(\tilde{\beta}|X) &= E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)'] \\ &= (A + B)Euu'(A + B)' \\ &= \sigma^2(X'X)^{-1} + \sigma^2BB' \\ &= V(\hat{\beta}|X) + \sigma^2BB' \end{aligned}$$

The last equation suggests that the variance of the estimator $\tilde{\beta}$ is larger than the variance of the estimator $\hat{\beta}$. This can be stated in the following important theorem,

Theorem 1. (Gauss-Markov) If $E(u|X) = 0$ and $E(uu'|X) = \sigma^2I$ in a linear regression model, then the OLS estimator is efficient e.g. $V(\tilde{\beta}|X) \geq V(\hat{\beta}|X)$ or

$$V(\tilde{\beta}|X) - V(\hat{\beta}|X) \text{ is positive semi-definite}$$

Note that the theorem is for conditional variances. Can we claim that $V(\tilde{\beta}) \geq V(\hat{\beta})$ also? We need a bit more work ... Consider first the definition

$$V(\hat{\beta}) = E(V(\hat{\beta}|X)) + V(E(\hat{\beta}|X)).$$

First, note that the second term is equal to zero because the estimator is unbiased, and therefore, $V(\beta) = 0$. Thus,

$$V(\hat{\beta}) = E(V(\hat{\beta}|X)) = \sigma^2 E((X'X)^{-1}).$$

Similarly,

$$V(\tilde{\beta}) = E(V(\tilde{\beta}|X)) = \sigma^2 [E((X'X)^{-1}) + E(BB')].$$

Therefore, we conclude that

$$V(\tilde{\beta}) \geq V(\hat{\beta}),$$

because $E(BB')$ is positive semi-definite.

3 Projections

Define the projection matrix

$$P = X(X'X)^{-1}X'.$$

The i th diagonal element of P is labeled leverage point and is defined as $h_i = x_i'(X'X)^{-1}x_i$. The matrix P is symmetric and idempotent, and its trace is equal to the number of independent variables. Symmetry implies,

$$P' = (X(X'X)^{-1}X')' = (X')'((X'X)^{-1})'(X)' = X(X'X)^{-1}X' = P.$$

Idempotency implies,

$$PP = X(X'X)^{-1}X'X(X'X)^{-1}X' = X(X'X)^{-1}X' = P.$$

Lastly,

$$\text{tr}(P) = \text{tr}(X(X'X)^{-1}X') = \text{tr}(X'X(X'X)^{-1}) = \text{tr}(I_p) = p.$$

The vector of fitted values, defined as

$$\hat{y} = X\hat{\beta},$$

represents the orthogonal projection of y onto the column space of X . Note that,

$$\hat{y} = X\hat{\beta} = X(X'X)^{-1}X'y = Py$$

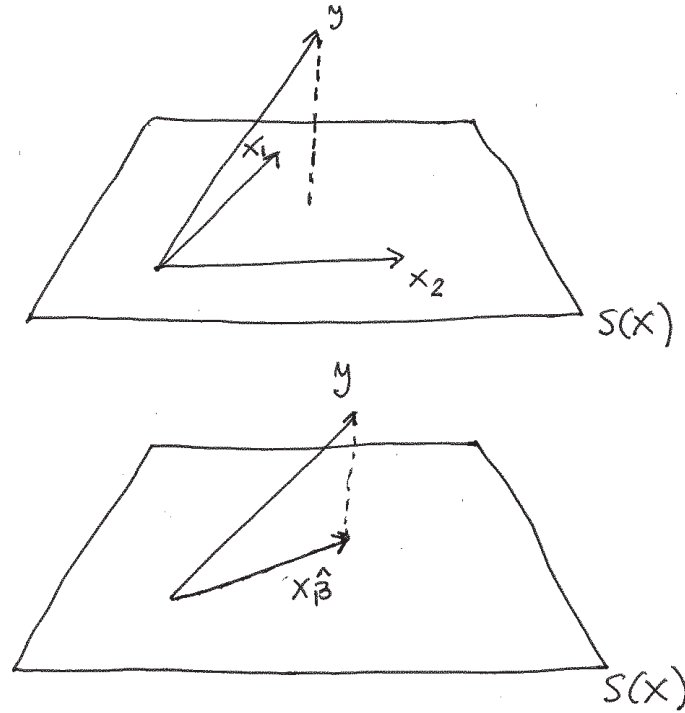


Figure 1: OLS fitting and projections

where the matrix P is the projection matrix. On the other hand, the residual vector

$$\hat{u} = y - \hat{y} = (I - P)y = My,$$

where $M := I - P$, which is a matrix that produces ‘residuals’. Note that we can also write the vector of residuals as

$$\hat{u} = My = M(X\beta + u) = MX\beta + Mu = Mu$$

because $PX = X$. This implies that the residuals from the regression and the independent variables are orthogonal,

$$X'\hat{u} = X'Mu = X'(I - P)u = (X' - X'P)u = 0.$$

Remark 1. The basic geometry of OLS can be seen in Figure 1. Let y and $X = [X_1 : X_2]$ be vectors, and define the space spanned by the columns of X as:

$$S(X) := \{z \in R^n | z = Xc, \quad c \in R^p\}$$

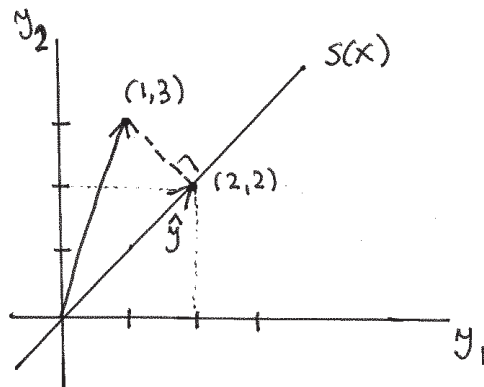


Figure 2: OLS fitting with two observations

The OLS fitting gives the closest vector to y in the space spanned by the X s.

Example 1. To illustrate the basic geometry of OLS, we present a simple example (Figure 2). Let $y = (1, 3)'$ and $X = (1, 1)'$ be vectors, and consider the regression model $y_i = \beta X_i + u_i$. The OLS estimator is

$$\arg \min \{(y_1 - \beta)^2 + (y_2 - \beta)^2\}$$

and the associated first order condition,

$$-2(y_1 - \beta) - 2(y_2 - \beta) = 0, \Rightarrow \bar{y} = \frac{1}{2} \sum y_i = 2.$$

Therefore, we obtain $\hat{y} = \hat{\beta}X = (2, 2)'$ and $\hat{u} = y - \hat{y} = (-1, 1)'$ with $X'\hat{u} = 0$. As it is clear in the figure, it is also interesting to obtain the length of the following vectors using the Euclidean norm:

$$\begin{aligned} \|y\| &= \sqrt{1^2 + 3^2} = \sqrt{10}, \quad \|\hat{y}\| = \sqrt{2^2 + 2^2} = \sqrt{8}, \\ \|u\| &= \|y - \hat{y}\| = \sqrt{(y_1 - 2)^2 + (y_2 - 2)^2} = \sqrt{2}. \end{aligned}$$

4 Partitioned Fit (Frish-Waugh-Lovell Theorem)

Consider a partitioned regression model,

$$y = X\beta + Z\gamma + u, \tag{5}$$

and the corresponding estimator $\hat{\beta}$. Moreover, consider the alternative regression model,

$$\tilde{y} = \tilde{X}\beta + \tilde{u}, \quad (6)$$

with the corresponding estimator $\tilde{\beta}$, with $\tilde{z} = M_Z z$ for a generic vector z .

Theorem 2. *The estimators $\hat{\beta}$ and $\tilde{\beta}$ are numerically identical.*

Proof: Standard OLS formulae in model (5) gives the following normal equations,

$$\begin{aligned} X'y &= X'X\hat{\beta} + X'Z\hat{\gamma} \\ Z'y &= Z'X\hat{\beta} + Z'Z\hat{\gamma} \end{aligned}$$

Solving for $\hat{\gamma}$ in the second equation, and replacing in the first equation, we obtain

$$X'y = X'X\hat{\beta} + X'Z((Z'Z)^{-1}Z'y - (Z'Z)^{-1}Z'X\hat{\beta}),$$

suggesting that

$$X'(I - Z(Z'Z)^{-1}Z')y = X'(I - Z(Z'Z)^{-1}Z')X\hat{\beta}$$

Therefore,

$$\hat{\beta} = (X'M_Z X)^{-1}X'M_Z y = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y} = \tilde{\beta}$$

■

In addition to the point estimates, the residuals are identical. To see this consider,

$$y = \hat{y} + \hat{u} = X\hat{\beta} + Z\hat{\gamma} + \hat{u}$$

and multiply by M_Z ,

$$M_Z y = M_Z X\hat{\beta} + M_Z Z\hat{\gamma} + M_Z \hat{u} \Rightarrow \tilde{y} = \tilde{X}\tilde{\beta} + M_Z \hat{u},$$

but $M_Z \hat{u} = \hat{u}$ because $P_Z \hat{u} = 0$.

Another way is seeing the this result is considering the following argument. Recall that

$$y = \hat{y} + \hat{u} = Py + My = X\hat{\beta} + Z\hat{\gamma} + My \quad (7)$$

Note that M is a projection matrix that uses $[X:Z]$. Applying partitioned formulae, one can show that $M = I - P_{XZ} = I - (P_Z + M_Z X(X'M_Z X)^{-1}X'M_Z)$.

If we multiply (7) by M_Z , we have

$$M_Z y = M_Z X\hat{\beta} + M_Z Z\hat{\gamma} + M_Z My \quad (8)$$

Therefore, we need to show that $M_Z M = M$ or, in words, that the last term in equations

(7) and (8) are equal. Note that

$$\begin{aligned}
 M_Z M &= (I - P_Z)(I - P_{XZ}) \\
 &= I - P_Z - P_{XZ} + P_Z P_{XZ} \\
 &= I - P_Z - P_{XZ} + P_Z(P_Z + M_Z X(X' M_Z X)^{-1} X' M_Z) \\
 &= I - P_Z - P_{XZ} + P_Z P_Z + P_Z M_Z X(X' M_Z X)^{-1} X' M_Z \\
 &= I - P_Z - P_{XZ} + P_Z + 0 \\
 &= I - P_{XZ} = M,
 \end{aligned}$$

which gives the results since $My = Mu = \hat{u}$.

Note too that in the case of $Z'X = 0$, the “off-diagonal” elements of the inverse matrix in the partitioned formula are zero, then $M = I - P_{XZ} = I - (P_Z + P_X)$. Again, we have that $P_Z(P_Z + P_X) = P_Z$, and the result continues to hold.

The theorem implies that the OLS estimates of the slope parameter β can be obtained as follows:

1. Regress y on Z and obtain the residuals $\tilde{y}$
2. Regress X on Z and obtain the residuals $\tilde{X}$
3. Regress $\tilde{y}$ on $\tilde{X}$, and obtain the OLS estimate $\hat{\beta}$.

Example 2. Consider a simple bivariate model,

$$y = \gamma + \beta x_i + u_i$$

where in this case, $X = x_i$ and $Z = \mathbf{1}$. Before turning to the application of FWL theorem, we can use the method of moments to derive the slope estimator. Our model assumptions were

$$\begin{aligned}
 E(u) &= E(y - \gamma - \beta x) = 0 \\
 \text{Cov}(x, u) &= E(xu) - E(x)E(u) = E(xu) = E(x(y - \gamma - \beta x)) = 0
 \end{aligned}$$

These two equations are restrictions imposed to the data about the joint probability distribution of x and y . There are two unknowns and two equations, which suggests that we can solve for γ and β . The sample counterpart of these equations are

$$\frac{1}{n} \sum_{i=1}^n (y_i - \gamma - \beta x_i) = 0 \quad (9)$$

$$\frac{1}{n} \sum_{i=1}^n (x_i(y_i - \gamma - \beta x_i)) = 0 \quad (10)$$

From equation (6), we obtain $\hat{\gamma} = \bar{y} - \hat{\beta}\bar{x}$, and replacing in (8), we obtain

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta} \sum_{i=1}^n (x_i - \bar{x})^2$$

which implies that the estimator for the parameter β is

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (11)$$

Following the FWL theorem, we first have to obtain the residuals in the regression,

$$x_i = \alpha + \epsilon_i \Rightarrow \tilde{x}_i = x_i - \hat{\alpha} = x_i - \bar{x}$$

Similarly,

$$y_i = \delta + v_i \Rightarrow \tilde{y}_i = y_i - \hat{\delta} = y_i - \bar{y}$$

Finally, we have to regress the residuals of the response variable on the residuals of the variable of interest,

$$\tilde{y}_i = \pi \tilde{x}_i + w_i$$

giving,

$$\hat{\pi} = \frac{\sum \tilde{x}_i \tilde{y}_i}{\sum \tilde{x}_i^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \hat{\beta} \quad (12)$$

Let's check now that the residuals in the two equations are equal, so the standard errors of $\hat{\beta}$ and $\hat{\pi}$ are equal. Again, by definition,

$$y_i = \hat{\gamma} + \hat{\beta}x_i + \hat{u}_i, \quad (13)$$

and

$$\tilde{y}_i = \hat{\pi}\tilde{x}_i + \hat{w}_i. \quad (14)$$

We can write the last equation as,

$$y_i - \bar{y} = \hat{\pi}(x_i - \bar{x}) + \hat{w}_i = \hat{\beta}(x_i - \bar{x}) + \hat{w}_i$$

and therefore,

$$\begin{aligned} y_i &= (\bar{y} - \hat{\beta}\bar{x}) + \hat{\beta}x_i + \hat{w}_i, \\ &= \hat{\gamma} + \hat{\beta}x_i + \hat{w}_i, \end{aligned}$$

using the definition of the intercept $\hat{\gamma}$. (Recall that we discuss this in Lecture 1). Notice then that the fitted values are equal, and therefore, the residual in equation (13) is equal to

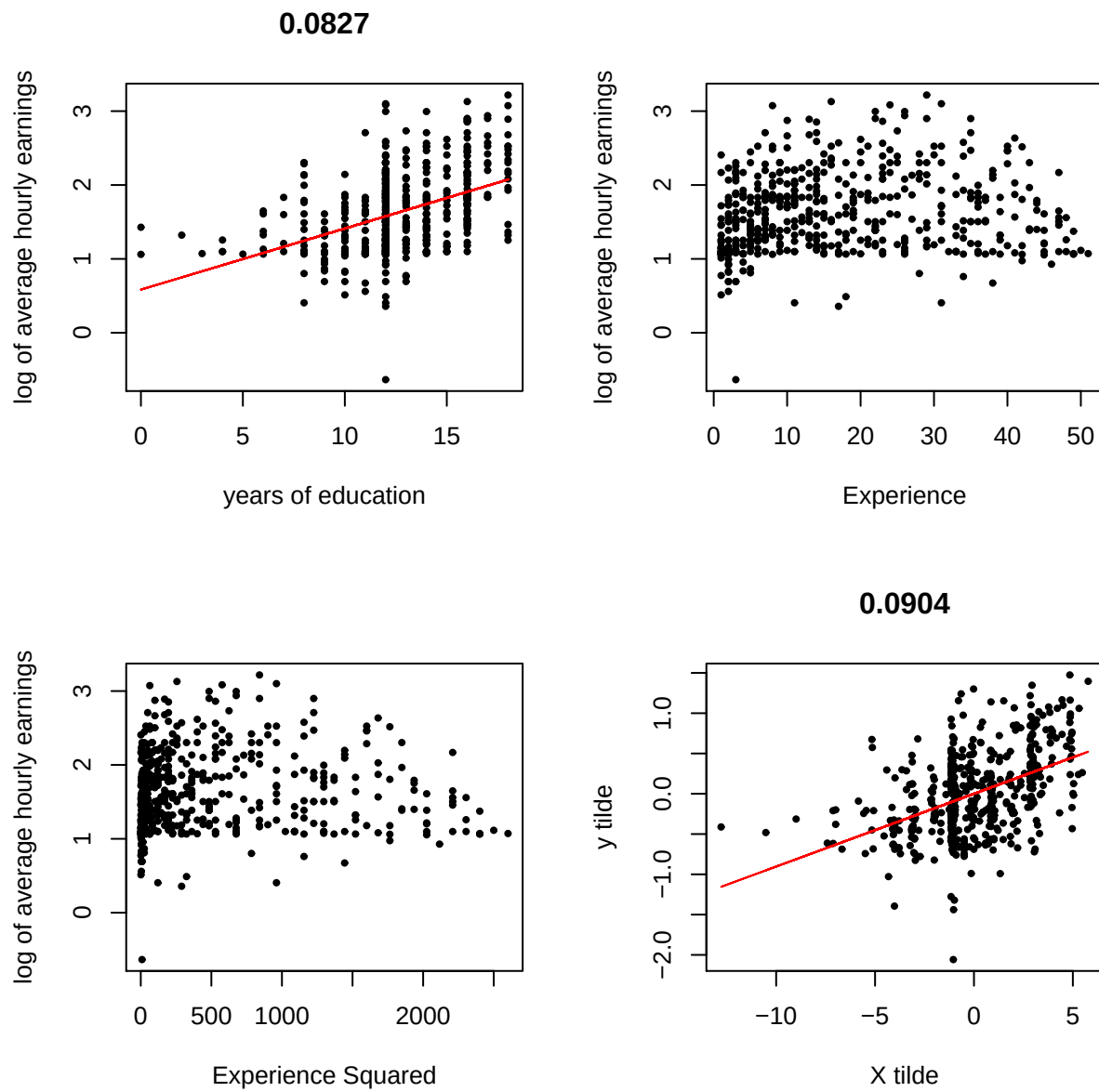


Figure 3: Partial Residual Plot

the residual in equation (14).

Example 3. Suppose we want to estimate the returns to education considering the following model,

$$lw_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i \quad (15)$$

where lw_i is the log of wages of the i^{th} worker, and x_{1i} , x_{2i} , and x_{3i} are the level of education, year of experience, and year with the actual employer of the i^{th} worker. The variable of interest is education, therefore we can rewrite the model using our previous notation:

$$y = Z'\gamma + \beta X + u$$

where $y = [lw]$, $Z = [x_2, x_3]$ and $X = [x_1]$.

We use a Wooldridge's (2002) data set to estimate the previous model. The data set contains observations for 500 individuals (Figure 1 shows the bivariate relation between dependent and independent variables). The last plot shows the slope obtained using partial residual plot. The return to education is 9 percent with an standard error of 0.0075.

5 Misspecification and Trade-offs

Under the classical assumptions the OLS estimator is unbiased, e.g. $E(\hat{\beta}) = \beta$. However, the estimator may be biased if we omit a variable in the regression model. Introducing variables in the model to 'correct' for the bias problem does not work because the variance of the estimators from the 'long' version of the model (e.g., the model that includes the additional explanatory variable) may be higher than the variance of the estimators from the 'short' version of the model (e.g., the model that does not include the explanatory variable).

Under the assumptions discussed so far, consider:

$$\begin{aligned} y_i &= x_i' \beta + z_i' \gamma + u_i \\ y_i &= x_i' b + v_i. \end{aligned}$$

Note that the second equation gives a projection of y on x_1 . In this case, we have:

$$\begin{aligned} b &= (Ex_i x_i')^{-1} (Ex_i y_i) = (Ex_i x_i')^{-1} (Ex_i (x_i' \beta + z_i' \gamma + u_i)) \\ &= (Ex_i x_i')^{-1} (Ex_i x_i') \beta + (Ex_i x_i')^{-1} (Ex_i z_i') \gamma + (Ex_i x_i')^{-1} (Ex_i u_i) \\ &= \beta + (Ex_i x_i')^{-1} (Ex_i z_i') \gamma, \end{aligned}$$

which obviously implies that the projection in the second equation does not identify a coefficient of interest β unless $Ex_i z_i' = 0$.

Let's discuss now the trade-off we often face in applied econometrics when we do not

observe a few independent variables. Suppose now that the true model is,

$$y = X\beta + Z\gamma + u, \quad (16)$$

but instead we estimate model,

$$y = X\beta + \epsilon. \quad (17)$$

The OLS estimator of the parameter of interest β ,

$$\begin{aligned} \hat{\beta}_s &= \arg \min \|y - X\beta\|^2, \\ &= (X'X)^{-1}X'y, \end{aligned}$$

could be biased if either $\gamma \neq 0$ or the observables independent variables X and Z are not independent,

$$E(\hat{\beta}_s) = \beta + E((X'X)^{-1}X'(Z\gamma + u)) = \beta + \Gamma\gamma$$

with $\Gamma = (X'X)^{-1}X'Z$. Note that Γ is the estimator of a regression of Z on X .

Example 4. We start studying the bias effect from omitting a relevant variable in the regression model. Consider that the true population model is,

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + u \quad (18)$$

but we obtain OLS estimators from a model like,

$$y = \beta_0 + \beta_1x_1 + u, \quad (19)$$

which excludes the relevant variable x_2 . Suppose we are interested in the slope parameter β_1 . In this case,

$$\begin{aligned} E(\tilde{\beta}_1) &= \beta_1 + \frac{\beta_2 \sum_{i=1}^n (x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \\ &= \beta_1 + \beta_2\tilde{\alpha}_1 \end{aligned}$$

What is the interpretation of $\tilde{\alpha}_1$? Note that if we have a simple regression model of the form,

$$x_2 = \alpha_0 + \alpha_1x_1 + \epsilon,$$

and we obtain the OLS estimator for the slope parameter α_1 , this is equal to

$$\tilde{\alpha}_1 = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

Therefore, the omitted variable bias is

$$E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\alpha}_1 \quad (20)$$

The bias is zero if either $\beta_2 = 0$, which means that the independent variable x_2 does not have an effect on the dependent variable y , or $\tilde{\alpha}_1 = 0$, which means that the independent variables x_1 and x_2 are uncorrelated. If $\beta_2 \approx 0$ or $\tilde{\alpha}_1 \approx 0$, the size of the bias would be small, and therefore it may not be a cause of concern.

Sometimes it is possible to infer the sign of the bias. If the sign of the unknown parameter β_2 is equal to the sign of the correlation between the independent variables x_1 and x_2 , the bias is positive. In this case, we say that $\hat{\beta}_1$ is upward bias. On the other hand, if the sign of β_2 is different than the sign of the correlation between the independent variables, the bias is negative. It is said that in this case, the estimator $\tilde{\alpha}_1$ is downward bias.

Example 5. Consider that y is log of wages, x_1 denotes years of education, and x_2 is ability or ‘spunk’ of some workers. We may believe that β_2 is positive and different from zero because there are evidence that suggests that more talented people receive higher wages. On the other hand, we suspect that $\text{Corr}(x_1, x_2) \neq 0$. The talented people tends to spend more years in school, so we infer that the relationship between education and ability is positive. Consequently the bias may be positive, so it may be the case that the OLS estimate of β_1 .

Although $\hat{\beta}_s$ is biased, it is efficient compared to the estimator of β_1 from the long model,

$$\hat{\beta}_l = (X' M_z X)^{-1} (X' M_z y)$$

To see this, first write the estimator from a short model as,

$$\hat{\beta}_s = (X' X)^{-1} X' (\hat{y}_l + \hat{u})$$

where $\hat{y}$ and $\hat{u}$ are the vectors of fitted values and residuals from the long model. Thus,

$$\hat{\beta}_s = \hat{\beta}_l + \Gamma \hat{\gamma}$$

Thus the variance of the estimator in the long model is

$$V(\hat{\beta}_l) = V(\hat{\beta}_s - \Gamma \hat{\gamma}) = V(\hat{\beta}_s) + \Gamma V(\hat{\gamma}) \Gamma', \quad (21)$$

which implies that

$$V(\hat{\beta}_l) \geq V(\hat{\beta}_s),$$

since the quadratic term is positive definite. The result in equation (21) holds if the (missing) covariance term is zero. We verify the conditions under which equation (21) holds in the following theorem:

Theorem 3. $\text{Cov}(\hat{\beta}_s, \Gamma \hat{\gamma}) = 0$

Proof: First, recall that

$$\begin{aligned}\hat{\gamma} &= (Z'M_xZ)^{-1}(Z'M_xy) \\ &= \gamma + (Z'M_xZ)^{-1}(Z'M_xu).\end{aligned}$$

Also, recall that

$$\begin{aligned}\hat{\beta} &= (X'M_zX)^{-1}(X'M_zy) \\ &= \beta + \Gamma\gamma + (X'M_zX)^{-1}(X'M_zu).\end{aligned}$$

By definition, we have that

$$\begin{aligned}Cov(\hat{\beta}_s, \Gamma\hat{\gamma}) &= E \left\{ \left(\hat{\beta}_s - E(\beta_s) \right) (\hat{\gamma} - E(\hat{\gamma}))' \Gamma' \right\} \\ &= E \left\{ \left(\hat{\beta}_s - E(\beta_s) \right) (\hat{\gamma} - \gamma)' \Gamma' \right\} \\ &= E \left\{ (X'X)^{-1} X'u ((Z'M_xZ)^{-1} ZM_xu)' \Gamma' \right\},\end{aligned}$$

by replacing by definitions. The last expression can be written as

$$\begin{aligned}Cov(\hat{\beta}_s, \Gamma\hat{\gamma}) &= E \left((X'X)^{-1} X'uu' M_x Z (Z'M_xZ)^{-1} \Gamma' \right) \\ &= \sigma^2 \left((X'X)^{-1} X' M_x Z (Z'M_xZ)^{-1} \Gamma' \right)\end{aligned}$$

because $V(u) = \sigma^2 I$. Note that $M_x X = 0$, therefore, covariance is equal to zero under homocedasticity. ■

We now summarize the main points. Since in applied economics usually both (a) the independent variables are correlated, and (b) there are variables that are omitted since there are difficult to measure,

1. Omitting independent variables introduce bias,
2. Adding independent variables inflate variance.

Remark 2. Suppose now that we are interested on the effect of omitting a variable that is correlated with X_1 . Suppose now that there is matrix of variables Z and, consider decomposing X as,

$$X\beta = X_1\beta_1 + X_2\beta_2 = P_z X_1\beta_1 + M_z X_1\beta_1 + X_2\beta_2$$

For $\gamma = 0$, we have that

$$\begin{aligned}X\beta &= P_z X_1(\beta_1 + \gamma) + M_z X_1\beta_1 + X_2\beta_2 \\ &= X_1\beta_1 + X_2\beta_2 + P_z X_1\gamma\end{aligned}$$

Therefore, we can now use an F test on $H_0 : \gamma = 0$ using the following regression

$$y = X\beta + \hat{X}_1\gamma + u$$

This is related to the auxiliary regression version of the celebrated Hausman (1978) test (see Wu (1973) also). Note that if you replace $\hat{X}_1$ by a matrix containing powers of $\hat{y}$ (e.g., $\hat{y}^2$, $\hat{y}^3$, etc.), we have Ramsey's test for specification error.

6 Practical Implications

If X and Z in equation (16) are not correlated, then there is no omitted variable bias and, if the PRL is correctly specified, the parameter β_1 in (17) can have a causal interpretation. Experiments however are not common, so researchers rely on control variables to eliminate biases.

This is connected to what is known as the *conditional independence assumption*, or CIA (as in Angrist and Pischke 2009). This is a condition we want to satisfy for estimation and inference and here is why. Consider a simple bivariate model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where β_1 is a constant causal parameter or effect. For instance, y_i is earnings and x_i is having a college degree or an indicator of veteran status. The error term ϵ_i capture other factors affecting earnings. If,

$$\epsilon_i = \delta_0 + \delta_1 z_i + v_i,$$

where z_i denote an “observable” variable or simply a vector of observables. Because v_i is assumed to not be correlated with z_i , we have that $E(\epsilon_i|x_i) \neq 0$ but $E(v_i|x_i, z_i) = 0$. Therefore, we estimate an “augmented” model with control z_i ,

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + u_i,$$

by OLS, obtaining an unbiased estimate of β_1 . Note that this was not possible if we use OLS to estimate (6).

Example 6. The role of covariates can be seen in the illustration presented in Table 1. We use Social Security and administrative data from Angrist (1998) which studied the effect of voluntary military service on earnings of soldiers. We use workers who applied to the military between 1976 and 1981, and we estimate the model before military service and after (i.e., years 1974 and 1985, as shown in the table). Voluntary military service is obviously non-random and most likely correlated with education, age, test scores, cohort effects, etc. We also know that these observables affect earnings. Therefore, omission of these variables

Table 1: The effect of veteran status on life-cycle earnings by race

	Log earnings (non-white)				Log earnings (white)			
	74	74	85	85	74	74	85	85
Veteran	0.002 (0.074)	-0.034+ (0.018)	0.131** (0.012)	0.085** (0.007)	-0.079 (0.072)	-0.049** (0.018)	0.040** (0.010)	-0.003 (0.006)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,763	1,763	2,499	2,499	2,096	2,096	3,155	3,155
R-squared	0.000	0.942	0.048	0.688	0.001	0.942	0.005	0.664
N	1763	1763	2499	2499	2096	2096	3155	3155

Standard errors in parentheses

** p<0.01, * p<0.05, + p<0.1

are likely to introduce biases.

The point is that after controlling for all these “observable” differences (using interactions too), veterans and non-veterans are comparable. Whether you believe of not this premise, the results shown in Table 1 points to the fact that the model estimated without controls give results that appear misleading.

Remark 3. We refer to controls as exogenous variables or observables, not to variables that are determined by the variable of interest. These controls are not potentially right hand side variables for other equations (or endogenous variables in genera). For instance, in the Mincer equation,

$$\log(wages)_i = \beta_0 + \beta_1 educ_i + \alpha_i + u_i,$$

α_i measures ability of the worker i and it is not observed (part of the error term). It makes sense to assume that $\text{Cov}(\alpha_i, educ_i) \neq 0$. In the CIA world, it is tempting to include variables to satisfy the condition and, sometimes, these variables are available to practitioners. For instance, you might have test scores used to screen job applicants. The real problem is that the performance in the test is likely to be correlated with education too.

For simplicity, consider that test scores are modeled as $score_i = \pi_0 + \pi_1 educ_i + \alpha_i + \varepsilon_i$. If we use test scores as a control variable, we estimate

$$\log(wages)_i = \gamma_0 + \gamma_1 educ_i + \gamma_2 score_i + v_i,$$

where $\gamma_1 = \beta_1 - \pi_1 \neq \beta_1$. Angrist and Pischke (2009) call these variables, including the test score used in this example, ‘bad controls’.

Large Sample Theory

Lecture 3

Our maintained assumptions so far have been:

- A1 The model is linear and given by $y_i = x_i'\beta + u_i$.
- A2 The rank of the matrix X is p .
- A3 The conditional expected value of y is equal to $E(y_i|X) = x_i'\beta$ for all subjects i in the sample of size n .
- A4 The conditional variance of y is equal to $V(y|X) = V(X\beta + u|X) = V(u) = \sigma^2 I$, where $\sigma^2 > 0$ and I is the identity matrix.

It is often assumed that $u_i \sim \mathcal{N}(0, 1)$, but this condition is too strong. We can replace it by a weaker condition on the moments of the random variables:

- A5 The observations $\{(y_i, x_i); i = 1, 2, \dots, n\}$ are *i.i.d.* across i and their fourth moments exist, i.e., $E(z_i^4) < \infty$ for a generic variable z . Also $E(x_i x_i')$ is a non-singular matrix.

We assume that y_i and x_i are jointly distributed and their fourth moments exists (e.g., $E(z^4) < \infty$ for $z = \{x, y\}$). An implication is that the error term u_i does not need to be normally distributed but their fourth moment should exist: $E(u^4) < \infty$.

1 Asymptotics as a tool

Several results obtained in econometrics are based on the normality assumption. This is certainly not a mild assumption and does not hold in practice.

Asymptotics is a tool that helps us to approximate the distribution of estimators or test statistics without assuming a parametric distribution, provided of course that the number of observations n (i.e. sample size) is sufficiently ‘large’. Most econometric methods used in applied economics perform well when the sample size is large. Moreover, sometimes is very difficult to obtain the standard error of non-linear estimators in linear models as in the following examples. We use an asymptotic approximation of the distribution of the estimator called ‘delta method’ (see Lemma 3.9 in Wooldridge 2010 and Section 5 below).

Example 1. Consider the model $y_t = x_t'\beta + u_t$ with t denoting time, and the interest is in estimating the elasticity parameter, $\eta_j = \frac{\partial E(y|x)}{\partial x_j} \frac{x_j}{E(y|x)}$. An estimator is,

$$\hat{\eta}_j = \hat{\beta}_j \frac{\bar{x}_j}{\hat{y}},$$

but the standard error of $\hat{\eta}_j$ is difficult to obtain because $\hat{y}$ and $\bar{x}_j$ are random variables. We can apply the delta method and obtain the standard error of $\hat{\eta}_j$. (We need to apply chain rule since $\beta_j \frac{x_j}{x'\beta}$).

Example 2. Consider a dynamic model of the form, $y_t = \gamma y_{t-1} + x_t'\beta + u_t$, with variables in logs. The long run elasticity of a change in x_j is, in this case, $\eta_j = \beta_j / (1 - \gamma)$. Again, the estimator is non-linear in parameters,

$$\hat{\eta}_j = \frac{\beta_j}{1 - \hat{\gamma}},$$

so the standard error of $\hat{\eta}_j$ can be obtained by applying the delta method.

2 Background

2.1 Large sample results

Consider the sequence of random variables $\{X_n : n = 1, 2, \dots\}$ and let X be a real valued random variable on a well defined probability space.

Definition 1 (Convergence in probability). We say that X_n converges in probability to X if

$$P(|X_n - X| > \epsilon) \rightarrow 0, \quad \text{as } n \rightarrow \infty \quad (1)$$

for any $\epsilon > 0$.

Intuitively, the probability that the difference $|X_n - X|$ takes a large value is very small as the sample size n increases. If $X_n \rightarrow \beta$, where β is a constant, we say that X_n is consistent and, often, we denote it by $\text{plim}_{n \rightarrow \infty} X_n = \beta$.

Why convergence concepts are important? We need a notion of “closeness” to unknown parameters. Recall that we have an i.i.d. (independently and identically distributed) random sample $X_1, X_2, \dots$ with realizations $x_1, x_2, \dots$. The joint distribution can be obtained by multiplying the marginal distribution. Suppose also that we are interested in two parameters, the population mean $E(X) = \mu$ and population variance, $\text{Var}(X) = \sigma^2$. We can estimate

them with,

$$\begin{aligned}\bar{X}_n &= T_x(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i \\ S^2 &= T_s(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.\end{aligned}$$

How “close” $\bar{X}_n$ is to μ and S^2 is to σ^2 ? If the sample size is sufficiently large, we expect $(\bar{X}_n, S^2)$ to be close to (μ, σ^2) . Now suppose we have a random experiment and we obtain basic outcomes s from the sample space S . We observe $x_i = X_i(s)$ and we can obtain

$$\bar{x}_n = \bar{X}_n(s). \quad (2)$$

Note that different outcomes will create different real numbers $\{\bar{x}_n = \bar{X}_n(s) : n = 1, 2, \dots\}$. Figure 1 illustrates the performance of $\bar{X}_n(s)$ and $S^2(s)$ for 3 different outcomes s_1, s_2 and s_3 from S .

Example 3. Suppose $Y_1, Y_2, \dots, Y_n$ is an i.i.d. random sample from a uniform distribution $\mathcal{U}(0, \theta)$, where θ is an unknown parameter. It is natural to estimate θ considering the largest order statistic, i.e. $X_n = \max_{1 \leq i \leq n} \{Y_i\}$. Is this estimator consistent? Note that, for a small constant $\epsilon > 0$, we have that $\{|X_n - \theta| > \epsilon\} = \{X_n - \theta > \epsilon\} \cup \{X_n - \theta < -\epsilon\}$. Therefore,

$$\begin{aligned}P(|X_n - \theta| > \epsilon) &= P(X_n - \theta > \epsilon) + P(X_n - \theta < -\epsilon) = P(X_n > \theta + \epsilon) + P(X_n < \theta - \epsilon) \\ &= P(X_n < \theta - \epsilon) = P\left(\max_{1 \leq i \leq n} \{Y_i\} < \theta - \epsilon\right) \\ &= P(Y_1 < \theta - \epsilon, Y_2 < \theta - \epsilon, \dots, Y_n < \theta - \epsilon) = \prod_{i=1}^n P(Y_i < \theta - \epsilon) \\ &= \left(\frac{\theta - \epsilon}{\theta}\right)^n = \left(1 - \frac{\epsilon}{\theta}\right)^n \rightarrow 0, \text{ as } n \rightarrow \infty.\end{aligned}$$

Definition 2 (Almost sure convergence). We say that X_n converges with probability 1, or almost surely, to X if

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1 \quad (3)$$

Definition 3 (Convergence in q^{th} mean). We say that X_n converges in q^{th} mean to X if,

$$\lim_{n \rightarrow \infty} \mathbb{E}|X_n - X|^q = 0 \quad (4)$$

Definition 4 (Convergence in distribution). We say that X_n converges in distribution to X if

$$\lim_{n \rightarrow \infty} F_n(X) = F(X). \quad (5)$$

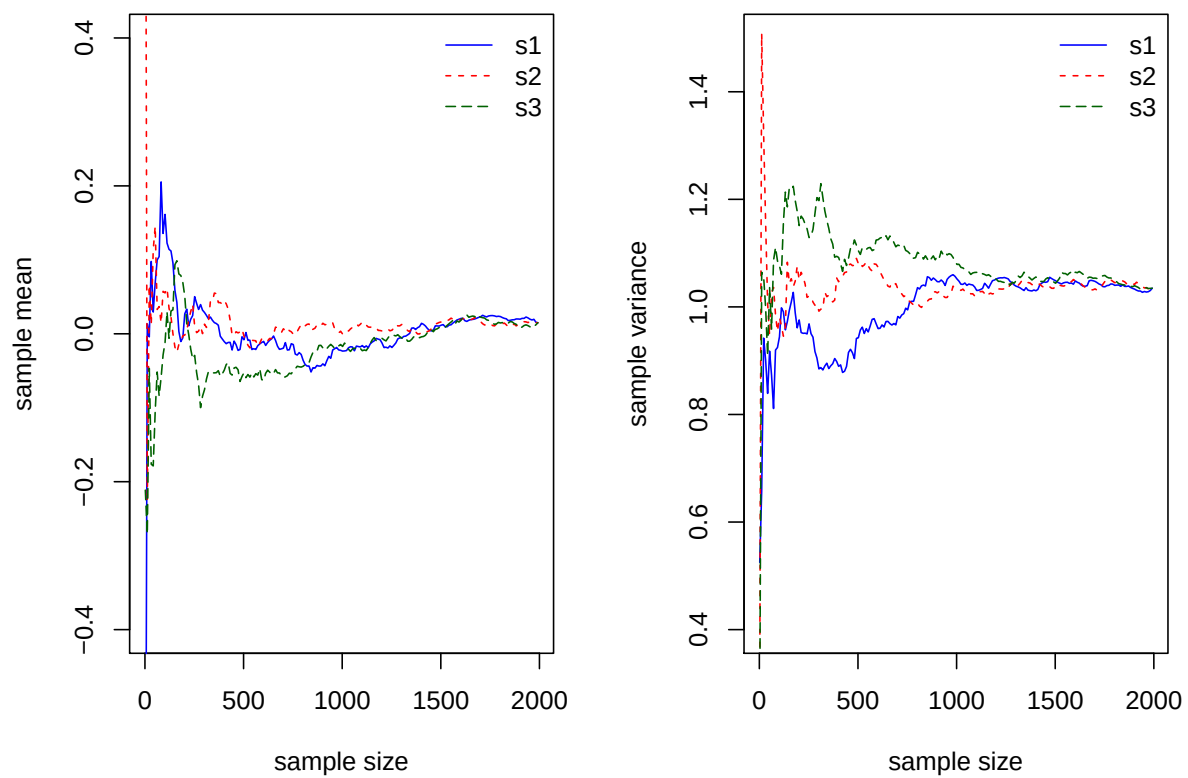


Figure 1: Sample mean and variance as sample size increases defined on a series of basic outcomes s

At each continuity point of F , we denote this as $X_n \xrightarrow{d} X$.

Theorem 1. *If $X_n \xrightarrow{\text{as}} X_1$, then $X_n \xrightarrow{P} X$, which says that almost sure convergence implies convergence in probability.*

Theorem 2. *If $X_n \xrightarrow{\text{rth}} X$, then $X_n \xrightarrow{P} X$*

Proof:

$$\mathbb{E}|X_n - X|^r \geq \mathbb{E}\{|X_n - X|^r I(|X_n - X| > \epsilon)\} \geq \epsilon^r P(|X_n - X| > \epsilon) \quad (6)$$

Therefore,

$$P(|X_n - X| > \epsilon) \leq \epsilon^{-r} \mathbb{E}|X_n - X|^r \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (7)$$

■

Corollary 1. *Convergence in probability, $X_n \xrightarrow{P} X$, implies convergence in distribution, $X_n \xrightarrow{d} X$.*

Corollary 2. *Convergence in probability, to a constant is equivalent to convergence in distribution to a given constant*

Theorem 3 (Asymptotic equivalence). *Consider a random sample (Y_n, X_n) with $Y_n \xrightarrow{d} y$, and $|Z_n| \xrightarrow{P} 0$, where $Z_n = Y_n - X_n$. It follows that $X_n \xrightarrow{d} y$.*

Proof: For $\epsilon > 0$, we can write $P(X_n < v) = P(Y_n < v + Z_n) = P(Y_n < v + Z_n, Z_n < \epsilon) + P(Y_n < v + Z_n, Z_n \geq \epsilon)$. Note too that $P(Y_n < v + Z_n, Z_n < \epsilon) \leq P(Y_n < v + \epsilon)$. Then it follows that

$$\begin{aligned} P(X_n < v) &\leq P(Y_n < v + \epsilon) + P(Y_n < v + Z_n, z_n \geq \epsilon) \\ &= P(Y_n < v + \epsilon) + [P(Z_n \geq \epsilon)P(Y_n < v + Z_n | z_n \geq \epsilon)] \\ &\leq P(Y_n < v + \epsilon) + P(Z_n \geq \epsilon) \end{aligned}$$

Note that we can similarly write, $P(X_n < v) = P(Y_n < v + Z_n) = P(Y_n < v + Z_n, Z_n > -\epsilon) + P(Y_n < v + Z_n, Z_n \leq -\epsilon)$ which leads to

$$P(X_n < v) \geq P(Y_n < v - \epsilon) + P(Z_n \geq -\epsilon).$$

By the assumption that $|Z_n| \xrightarrow{P} 0$, the last term tends to 0 as $n \rightarrow \infty$, so it can be ignored. We can write

$$P(Y_n < v - \epsilon) \leq P(X_n < v) \leq P(Y_n < v + \epsilon).$$

By picking ϵ arbitrarily small, we have that $F_X = F_Y$ as $n \rightarrow \infty$. ■

We can interpret this result as showing the asymptotic equivalence when two random variables are very close to each other. As an illustrative example, consider $Y_n = \sqrt{n}\bar{X}_n/S_n$

and $Z_n = \sqrt{n}\bar{X}_n/\sigma$, where σ is an unknown parameter. Dealing with a random denominator S_n is complicated, but we can first show that $Y_n - Z_n \xrightarrow{p} 0$ and then obtain the distribution of Y_n using the result that $Y_n \xrightarrow{d} Z$.

Theorem 4 (Slutsky). *Let $X_n \xrightarrow{d} x$ and $Y_n \rightarrow y$, a real constant. Then,*

1. $X_n + Y_n \xrightarrow{d} x + y$
2. $X_n Y_n \xrightarrow{d} yx$
3. $X_n/Y_n \xrightarrow{d} x/c$ if $c \neq 0$

Theorem 5 (Continuous Mapping). *If $X_n \xrightarrow{d} X$ and g is continuous, $g(X_n) \rightarrow g(X)$*

Example 4. If $X_n \xrightarrow{d} \mathcal{N}(0, 1)$, then $X_n^2 \xrightarrow{d} \chi_1^2$

Example 5. If $(X_n, Y_n) \xrightarrow{d} \mathcal{N}(0, I_2)$, then $X_n/Y_n \xrightarrow{d} z$, a Cauchy random variable.

Example 6. When g is not continuous, we may need to be careful. Consider the following example:

$$g(t) = \begin{cases} t - 1 & t \leq 0 \\ t + 1 & t > 0 \end{cases}$$

Let $X_n = 1/n$, then $X_n \rightarrow X = 0$, $g(X_n) \rightarrow 1$ but $g(X) = -1$.

2.2 Order relations and rates of convergence

We now examine order relations and rates of convergence. In this section, the terms $O(\cdot)$ ('big oh') and $o(\cdot)$ ('small oh') are used to determine the order of magnitude of deterministic and stochastic sequences.

Consider positive deterministic sequences $\{a_n\}$ and $\{b_n\}$.

Definition 5. $a_n = o(b_n)$ as $n \rightarrow \infty$ if $a_n/b_n \rightarrow 0$

Perhaps it is good now to have a more comprehensive definition.

Definition 6 (Orders of magnitude). A sequence $\{a_n\}$ is of order smaller than n^λ , which is denoted by $a_n = o(n^\lambda)$ or $n^{-\lambda}a_n = o(1)$, if for $n \geq N$, we have that

$$|n^{-\lambda}a_n| < \epsilon,$$

for every $\epsilon > 0$.

Definition 7 (Orders of magnitude). A sequence $\{b_n\}$ is *at most of order* n^λ , which is denoted by $b_n = O(n^\lambda)$ or $n^{-\lambda}b_n = O(1)$, if for $n \geq N$, we have that

$$|n^{-\lambda}b_n| < K,$$

where $K < \infty$ is a very large real number.

Remark 1. a_n and b_n tends to infinity, this says that a_n tends to infinity more slowly than b_n . If a_n and b_n tend to zero, this indicates that a_n tends to zero faster than b_n .

Example 7. $1/n^2 = o(1/n)$ as $n \rightarrow \infty$, because $(1/n^2)/(1/n) = n/n^2 = 1/n \rightarrow 0$

Remark 2. It makes a big difference whether a sequence tends to 0 at rate $1/n^2$ or $1/n$

Suppose

$$a_n = \frac{1}{n} - \frac{2}{n^2} + \frac{4}{n^3} \quad (8)$$

Let's approximate a_n by $b_n = 1/n$. Therefore,

$$a_n = \frac{1}{n} + R_n = \frac{1}{n} + o(1/n) \quad (9)$$

Where $R_n = \frac{-2}{n^2} + \frac{4}{n^3} = o(1/n)$. Or, if we approximate a_n by $1/n - 2/n^2$, we write

$$a_n = \frac{1}{n} - \frac{2}{n^2} + o(1/n^2) \quad (10)$$

Lemma 1. (a) $a_n = o(1)$ iff $a_n \rightarrow 0$;

(b) If $a_n = o(b_n)$ and $b_n = o(c_n)$, then $a_n = o(c_n)$;

(c) For any constant $c \neq 0$, $a_n = o(b_n)$ implies $ca_n = o(b_n)$;

(d) For any sequence of numbers $c_n \neq 0$, $a_n = o(b_n)$ implies $c_n a_n = o(c_n b_n)$;

(e) If $d_n = o(b_n)$ and $e_n = o(c_n)$, then $d_n e_n = o(b_n c_n)$

Example 8. Let $a_n = \log n$, and $b_n = n^k$. Then

$$a_n = o(b_n) \text{ for every } k > 0 \quad (11)$$

Definition 8. If a_n is of n order smaller than or equal to b_n , we say that

$$a_n = O(b_n) \quad (12)$$

If there is a $\Delta < \infty$, such that $a_n/b_n \leq \Delta$ for sufficiently large n , we say that $a_n = O(b_n)$. Note that a particular $a_n = O(1)$ iff the sequence a_n is bounded.

Lemma 2. *If $a_n = O(n^r)$ and $b_m = O(n^s)$, then:*

$$(a) a_n b_n = O(n^{r+s})$$

$$(b) a_n + b_n = O(n^{\max\{r,s\}})$$

The terms $O_p(\cdot)$ ('big oh') and $o_p(\cdot)$ ('small oh') are for stochastic sequences. Intuitively, we can say that a sequence $\{X_n : n = 1, 2, \dots\}$ is *of smaller order of magnitude* than n^λ if it converges in probability $X_n/n^\lambda \rightarrow 0$ as $n \rightarrow \infty$. It is denoted by $X_n = o_p(n^\lambda)$.

Similarly, we can say that a sequence $\{X_n : n = 1, 2, \dots\}$ is *at most of order* of magnitude n^λ if $P(|X_n/n^\lambda| > M) < \delta$, where $\delta > 0$. It is denoted by $X_n = O_p(n^\lambda)$, and it implies that X_n is bounded in probability.

Now consider sequences $\{X_n\}$ and $\{Y_n\}$ on the probability space $(\Omega, \mathcal{A}, \mathcal{P})$ and any $\epsilon > 0$,

Definition 9. a. If $\exists \Delta < \infty$ such that $P(|X_n| \geq \Delta|Y_n|) < \epsilon$ for sufficiently large n , we write

$$X_n = O_p(Y_n) \quad (13)$$

If Y_n is non stochastic, we say that X_n is bounded in probability and $X_n = O_p(1)$

b. If $P(|X_n| \geq \epsilon|Y_n|) \rightarrow 0$, we write $X_n = o_p(Y_n)$. We say that it is tending to zero in probability, i.e.,

$$\frac{X_n}{Y_n} \rightarrow 0. \quad (14)$$

3 A few LLNs and CLTs

The law of large numbers (LLN) and central limit theorem (CLT) plays a key role in econometrics. For an extensive discussion, please see White (2000) and Davidson (1994). In this section, we will review a few results with particular emphasis on the variations related to the distribution of a sequence of random variables.

Theorem 6 (Khinchine LLN). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_1) = \mu$. Then,*

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i - \mu \xrightarrow{p} 0. \quad (15)$$

Theorem 7 (Chebyshev LLN). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$. If $n^{-1} \sum_{i=1}^n \sigma_i^2 \leq K < \infty$ as $n \rightarrow \infty$, then,*

$$\bar{X}_n - \mu_n = \frac{1}{n} \sum_{i=1}^n \mu_i \xrightarrow{p} 0. \quad (16)$$

Theorem 8 (Kolmogorov LLN). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$. If $\sum_{i=1}^n \sigma_i^2/n^2 < \infty$ as $n \rightarrow \infty$, then $\bar{X}_n - \mu_n \rightarrow 0$ with probability 1 (wp 1).*

The literature on CLT is extensive, so I will only discuss a few results.

Theorem 9 (Lindberg-Levy CLT). *Let $\{X_i\}$ be a sequence of i.i.d. random variables with $\mathbb{E}(X_1) = \mu$ and $\text{Var}(X_1) = \sigma^2$. Then,*

$$\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1). \quad (17)$$

The Liapunov CLT and Linberg-Feller CLT cover the case of heterogeneously distributed random variables, while assuming independence. I introduce the following result which has a broader use in econometrics.

Theorem 10. *Let $\{X_i\}$ be a sequence of independent random variables with $\mathbb{E}(X_i) = \mu_i$, $\text{Var}(X_i) = \sigma_i^2$ and $E|X_i - \mu_i|^{2+\delta} < \infty$ for some positive δ . If $\lim_{n \rightarrow \infty} n^{-1} \sum_{i=1}^n \sigma_i^2 < \infty$, then,*

$$\sqrt{n}(\bar{X}_n - \bar{\mu}_n)/\bar{\sigma}_n \xrightarrow{d} \mathcal{N}(0, 1). \quad (18)$$

4 The Asymptotic Theory of OLS

4.1 Consistency

Definition 10. A consistent estimator $\hat{\theta}$ is one that converges to θ_0 in probability,

$$\hat{\theta} \rightarrow \theta_0, \quad (19)$$

as the sample size n goes to infinity, for all possible true values.

With this definition in mind, we now investigate consistency of OLS in the simplest conceivable setting. Consider a linear regression model of the form,

$$y_i = x_i' \beta + u_i, \quad (20)$$

where $E(u_i|x_i) = E(u_i) = 0$. We can tentatively assume that the error terms u are iid draws from F . The OLS estimator for the unknown parameter β is $\hat{\beta} = (X'X)^{-1}X'y$ and can be written as

$$\hat{\beta} - \beta = (X'X)^{-1}X'u \quad (21)$$

$$= \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i u_i \right) \quad (22)$$

where X is a $n \times p$ matrix and u is a $n \times 1$ vector. If the x 's are considered fixed or non-stochastic, we have that the first RHS term,

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \rightarrow Q_0 \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n x_i u_i \rightarrow 0.$$

Note that

$$\mathbb{E} \left(\frac{1}{n} \sum_{i=1}^n x_i u_i \right) = 0 \quad (23)$$

because $\mathbb{E}(u_i | x_i) = 0$ and the u 's are independent. Moreover,

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n x_i u_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(x_i u_i) \quad (24)$$

$$= \frac{1}{n^2} \sigma^2 \sum_{i=1}^n x_i x_i' \quad (25)$$

$$= \frac{\sigma^2}{n} \frac{1}{n} \sum_{i=1}^n x_i x_i' \rightarrow 0 \quad (26)$$

Then,

$$\hat{\beta} \rightarrow \beta, \quad (27)$$

follows by a simple application of Chebychev and the Slutsky theorem. (The case of stochastic x 's requires the use of SLLN, but the argument is similar).

Remark 3 (Inconsistency 1). Note that if $x_i = i$ in a model with no intercept and $p = 1$, then

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n i^2 \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n i u_i \right).$$

It is immediately apparent that

$$\frac{1}{n} \sum_{i=1}^n i^2,$$

does not converge to a constant, say Q_0 as before, and the inverse does not exist. Note that dividing by n^2 will not work either. However, if

$$n(\hat{\beta} - \beta) = \left(\frac{1}{n^3} \sum_{i=1}^n i^2 \right)^{-1} \left(\frac{1}{n^2} \sum_{i=1}^n i u_i \right),$$

the first term on the right hand side converges to $1/3$ as $n \rightarrow \infty$, and the variance of the

last term converges to zero.

Remark 4 (Super consistency). The super consistent estimator

$$n(\hat{\beta} - \beta) = O_p(n^{-3/2}),$$

while the standard OLS estimator when x_i is not a deterministic trend is

$$\hat{\beta} - \beta = O_p(n^{-1/2}).$$

Remark 5 (Inconsistency 2). If the main equation omits a variable, say variable v_i , then

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i (v_i + u_i) \right).$$

Following previous steps, we obtain

$$\hat{\beta} - \beta \rightarrow Q_0 Q_{xv} \neq 0,$$

where Q_{xv} is the covariance matrix of x_i and v_i and it is non-zero if the variables are correlated.

4.2 Asymptotic Normality

Consider (again) a linear regression model:

$$y_i = x_i' \beta + u_i \tag{28}$$

Recall that $\hat{\beta} - \beta = (X'X)^{-1}X'u$. We can write:

$$\begin{aligned} \sqrt{n}(\hat{\beta} - \beta) &= (X'X/n)^{-1} n^{-1/2} X'u \\ &= \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i \right) \end{aligned}$$

If we apply Corollary 2, we have that

$$\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i \right) \xrightarrow{d} \mathcal{N}(0, \sigma^2 Q_0) \tag{29}$$

By Slutsky theorem, we have that

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Q_0^{-1} \mathcal{N}(0, \sigma^2 Q_0) = \mathcal{N}(0, \sigma^2 Q_0^{-1}) \quad (30)$$

Note that we applied again the Slutsky theorem. In sum, asymptotic normality follows from (i) convergence in distribution of the score, (ii) convergence in probability of the Hessian, and (iii) the Slutsky theorem.

Remark 6. Key to the results derived before is the fact that,

$$\begin{aligned} S &= E(x_i x_i' u_i^2) = E\{E(x_i x_i' u_i^2 | x_i)\} \\ &= E\{x_i x_i' E(u_i^2 | x_i)\} = \sigma^2 E\{x_i x_i'\} = \sigma^2 Q_0, \end{aligned}$$

and if $E(u_i^2 | x_i) \neq \sigma^2$, we cannot simplify the expressions as in the last steps.

5 The Delta Method

For the method, the first result that we need is consistency: if the function g is nice and smooth, then $g(\hat{\theta}) \rightarrow g(\theta)$. Assuming that $\hat{\theta}$ is consistent and asymptotically normal with variance Ω , and that $g(\cdot)$ is smooth,

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \rightsquigarrow \mathcal{N}(0, G' \Omega G)$$

where

$$G = \nabla_{\theta} g(\theta)|_{\theta=\hat{\theta}} = \begin{pmatrix} \partial g(\theta)/\partial \theta_1 \\ \partial g(\theta)/\partial \theta_2 \\ \vdots \\ \partial g(\theta)/\partial \theta_p \end{pmatrix}$$

Why? Because Taylor expansion will claim that for $g(\hat{\theta})$ around $\hat{\theta} = \theta$,

$$g(\hat{\theta}) = g(\theta) + (\hat{\theta} - \theta) \nabla_{\theta} g(\hat{\theta})|_{\hat{\theta}=\theta} + (\theta^* - \theta)' \nabla_{\theta}^2 g(\hat{\theta})|_{\hat{\theta}=\theta} (\theta^* - \theta)$$

where $\theta_j^* \in (\hat{\theta}_j, \theta_j) \forall j$. Note that if the estimator $\hat{\theta}$ is consistent, then the 2nd order term vanishes asymptotically because

$$\hat{\theta}_i \rightarrow \theta_j \Rightarrow \theta_i^* \rightarrow \theta_j \quad \forall j$$

Therefore, asymptotically we have that

$$V(g(\hat{\theta})) = (\nabla_{\theta} g(\hat{\theta})|_{\hat{\theta}=\theta})' V(\hat{\theta}) (\nabla_{\theta} g(\hat{\theta})|_{\hat{\theta}=\theta})$$

6 Introduction to empirical processes

In modern econometrics, point-wise convergence is sometimes too weak to show that the estimator $\hat{\theta}$ is consistent for θ_0 . We use a concept that is related to “functional” convergence or uniform convergence of criterion functions. In this section, we will present an introduction to some results that are frequently used.

6.1 Background and definitions

Consider $X_1, \dots, X_n$ to be i.i.d. with distribution F defined F_n is a random function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n 1_{(-\infty, x]}(X_i),$$

for $X \in R$. Recall that for a fixed x , $F_n(x) \rightarrow F(x)$ by a LLN and

$$Z_n(x) = \sqrt{n}(F_n(x) - F(x)) \rightarrow Z \sim N(0, 1).$$

More general uniform convergence results on the empirical process are:

Theorem 11 (Glivenko - Cantelli (1933)).

$$\|F_n - F\|_{\infty} = \sup_x |F_n(x) - F(x)| \rightarrow 0$$

6.1.1 M - and Z - estimators

A popular method for finding estimators is to maximize a criterion function of the type

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n m_{\theta}(X_i),$$

For known functions $m_{\theta} : \chi \rightarrow R$. For instance, $m_{\theta}(x_i) = (x - \theta)^2$, $m_{\theta}(x_i) = |x - \theta|$ or $m_{\theta} = -\log P_{\theta}$. Consider the following definitions:

$$\begin{aligned} \theta_0 &= \arg \max_{\theta \in \Theta} M(\theta) \\ \hat{\theta}_n &= \arg \max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n m_{\theta}(X_i) = \arg \max_{\theta \in \Theta} M_n(\theta) \end{aligned}$$

Another way of thinking this problem is to consider, finding the solution of,

$$\Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n m'_\theta(X_i) = \frac{1}{n} \sum_{i=1}^n \Psi_\theta(X_i) = 0$$

The solution of this estimating equation is $\hat{\theta}$ are called Z -estimators. Note however, that the estimating equations may not need to correspond to a maximization problem.

6.2 Uniform consistency

We now investigate consistency for M estimators. Let $\hat{\theta}_n$ be an estimator of θ . We call it asymptotically consistent if the sequence $\hat{\theta}_n$ converges in probability for θ for every possible value of θ . (e.g. $\bar{X}$ for the population mean EX).

Let d be a metric space that includes the set of possible parameters. We want to prove that $d(\hat{\theta}_n, \theta_0) \rightarrow 0$.

The basic point of the issue of point-wise vs functional convergence, it that although

$$M_n(\theta) \rightarrow M(\theta) \quad \text{for every } \theta$$

and

$$\arg \max M_n(\theta) \rightarrow \arg \max M(\theta),$$

we cannot ensure convergence of $\hat{\theta}_n$. We need a “functional” convergence of M_n to M , to strength out the point-wise convergence. It is commonly used the notion of uniform convergence of the criterion functions.

Consider an estimator $\hat{\theta}_n$ that nearly maximize M_n :

$$M_n(\hat{\theta}_n) \geq \sup_{\theta \in \Theta} M_n(\theta) - o_p(1),$$

which of course implies that

$$M_n(\hat{\theta}) \geq M_n(\theta_0) - o_p(1)$$

Theorem 12. : For $\epsilon > 0$,

1. $\sup |M_n(\theta) - M(\theta)| \rightarrow 0$,
2. $\sup_{\theta: d(\theta, \theta_0) \geq \epsilon} M(\theta) < M(\theta_0)$

Then, $\hat{\theta}_n \rightarrow \theta_0$.

Proof: Consider that $\hat{\theta}_n$ satisfies $M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_p(1)$. By uniform convergence of M_n to M by Assumption 1, we have that $M_n(\hat{\theta}_n) \geq M(\theta_0) - o_p(1)$. Then,

$$\begin{aligned} M_n(\hat{\theta}_n) - M(\hat{\theta}_n) &\geq M(\theta_0) - M(\hat{\theta}_n) + o_p(1) \\ \sup_{\theta} |M_n(\hat{\theta}_n) - M(\theta)| &\geq M(\theta_0) - M(\hat{\theta}_n) + o_p(1) \end{aligned}$$

Therefore,

$$M(\theta_0) - M(\hat{\theta}_n) \leq \sup_{\theta} |M_n(\theta) - M(\theta)| + o_p(1) \rightarrow 0$$

For every θ , we have, by Assumption 2, that

$$M(\theta) < M(\theta_0) - \eta.$$

Thus, the event $\{d(\hat{\theta}_n, \theta) \geq \epsilon\}$ is contained in the event $\{M(\hat{\theta}_n) < M(\theta_0) - \eta\}$. We can write,

$$P(d(\hat{\theta}_n, \theta) \geq \epsilon) \leq P(M(\hat{\theta}_n) - M(\theta_0) > \eta) \rightarrow 0,$$

as $n \rightarrow \infty$. ■

Remark 7. Uniform convergence of $M_n(\theta)$ required by the stochastic condition is equivalent to the set of functions $\{m_{\theta} : \theta \in \Theta\}$ being Glivenko, Cantelli. A simple set of sufficient conditions is that Θ be compact, $m_{\theta}(x)$ be continuous for each x , and the function is dominated by an integrable function.

6.3 Asymptotic Normality

Without loss of generality, let now consider the Z estimators instead of the M estimators. Let a sample $X_1 \cdots X_n$ from a distribution P , and let the “true” criterion function be of the form,

$$\begin{aligned} \Psi_n(\theta) &= \frac{1}{n} \sum_{i=1}^n \psi_{\theta}(X_i) \\ \Psi(\theta) &= E\psi_{\theta}. \end{aligned}$$

Assume that $\hat{\theta}_n$ is a zero of Ψ_n and converges in probability to a zero θ_0 in Ψ . We expand $\Psi_n(\hat{\theta}_n)$ using Taylor series around θ_0 :

$$0 = \Psi_n(\hat{\theta}_n) = \Psi_n(\theta_0) + (\hat{\theta}_n - \theta_0)\Psi'_n(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)^2\Psi''_n(\theta_n^*),$$

where $\theta_n^* \in (\theta_0, \hat{\theta}_n)$. Thus,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{-\sqrt{n}\Psi_n(\theta_0)}{\Psi'_n(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)\Psi''_n(\theta_n^*)}. \quad (31)$$

Assuming that $V_\psi = E\psi_{\theta_0}^2$ is finite and considering that $E\psi_{\theta_0} = 0$,

$$-\sqrt{n}\Psi_n(\theta_0) = -\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i) \rightsquigarrow N(0, V_\psi).$$

We now consider the denominator of equation (31). The first term $\Psi'_n(\theta_0) \rightarrow E\psi'_{\theta_0}$ by LLN. The second term is

$$o_p(1) \times \psi''_n(\theta_n^*) = o_p(1) \times O_p(1) = o_p(1),$$

because $\hat{\theta}_n - \theta_0 \rightarrow 0$. Therefore, using Slutsky theorem,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightsquigarrow \mathcal{N}(0, (E\psi'_{\theta_0})^{-2}V_\psi).$$

Remark 8. We assume that the function $\theta \rightarrow \psi_\theta(X)$ possesses two continuous derivatives with respect to θ , for every X . This is not always true (e.g., the median). The condition can be replaced by a Lipschitz condition.

Hypothesis Testing in the Linear Regression Model

Lecture 4

This lecture introduces classical tests for the linear regression model. To be consistent with the previous lectures, we assume that the model is:

$$y = X\beta + u, \quad (1)$$

where the response variable y is an n -dimensional vector, X is a $n \times p$ matrix, β is a p dimensional vector and u is a n dimensional error term. We are interested in inference on the parameter of interest, β .

Recall the following assumptions:

- A1 The model is linear and given by equation (1).
- A2 The rank of the matrix X is p .
- A3 The conditional expected value of y is equal to $E(y_i|X) = x_i'\beta$ for all subjects i in the sample of size n .
- A4 The conditional variance of y is equal to $V(y|X) = V(X\beta + u|X) = V(u) = \sigma^2 I$, where $\sigma^2 > 0$ and I is the identity matrix.

Assumption A1 implies that the model is correctly specified and A2 implies that the projection matrix $P = X(X'X)^{-1}X'$ exists. Note that this also implies that the OLS estimator, $\hat{\beta} = (X'X)^{-1}X'y$ exists too. A3 implies that the error term u and the independent variables are not correlated:

$$E(u_i|X) = E(y_i - x_i'\beta|X) = E(y_i|X) - x_i'\beta = x_i'\beta - x_i'\beta = 0$$

Note that this also implies that $E(u_i) = E(E(u_i|X)) = 0$ and the covariance between independent variables and error term is zero. Lastly, A4 implies that the error term u has constant variance, σ^2 , and u_i and u_j are not correlated.

We add one more assumption that will be relaxed in Lecture 5:

A5 The conditional distribution of y is normal with mean $X\beta$ and variance $\sigma^2 I$.

In this lecture, we will discuss how to evaluate different hypothesis on the coefficients in the following regression model,

$$y_i = x_i' \beta + u_i = \beta_0 + \sum_{j=1}^p x_{ij} \beta_j + u_i.$$

For instance, the hypotheses we might want to evaluate are,

$$\begin{aligned} H_0 : \quad & \beta_1 = 0 \\ H_0 : \quad & \beta_1 = \beta_2 \\ H_0 : \quad & \beta_p + \beta_{p-1} = c \\ H_0 : \quad & \beta_1 = \beta_2 = \dots = \beta_p = 0. \end{aligned}$$

1 Linear hypothesis testing

A general expression for linear hypothesis is of the following form:

$$\begin{aligned} H_0 : R\beta &= \delta, \\ H_a : R\beta &\neq \delta \end{aligned}$$

where H_0 is the null hypothesis, H_a is the alternative hypothesis, R is a known $r \times p$ matrix with rank r , and δ is a $r \times 1$ vector. Here r indicates the number of restrictions applied to p parameters.

Example 1. Suppose $\beta = (\beta_1, \beta_2, \beta_3)$ in equation (1). Therefore,

$$\begin{aligned} r_{11}\beta_1 + r_{12}\beta_2 + r_{13}\beta_3 &= \delta_1, \\ r_{21}\beta_1 + r_{22}\beta_2 + r_{23}\beta_3 &= \delta_2. \end{aligned}$$

Suppose we want to test $\beta_1 + \beta_2 = 1$ and $\beta_1 - \beta_3 = 0$. Then,

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}; \quad \delta = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Note that the rank of R is 2.

Example 2. The simplest conceivable case is $r = 1$, then $R = [1, 1, 0]$, $\delta = 1$, and then test

$$H_0 : \beta_1 + \beta_2 = 1$$

How we can test the previous null hypothesis? Under Assumption A5, we have that

$$\begin{aligned}\hat{\beta} &\sim \mathcal{N}(\beta, \sigma^2(X'X)^{-1}) \\ \hat{\beta} - \beta &\sim \mathcal{N}(0, \sigma^2(X'X)^{-1}) \\ \sigma(X'X)^{1/2}\hat{\beta} - \beta &\sim \mathcal{N}(0, I_p).\end{aligned}$$

It can be shown then that,

$$R\hat{\beta} \sim N(R\beta, \sigma^2 R(X'X)^{-1}R').$$

Therefore, in the case that $r = 1$,

$$\frac{R\hat{\beta} - R\beta}{\sqrt{\sigma^2 R(X'X)^{-1}R'}} \sim N(0, 1).$$

Under the null hypothesis, it is equal to

$$\frac{R\hat{\beta} - \delta}{\sqrt{\sigma^2 R(X'X)^{-1}R'}} \sim N(0, 1)$$

Note however that this test statistic is unfeasible because σ^2 is unknown. We replace σ^2 by $\hat{\sigma}^2$ defined as,

$$\hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{n - p}$$

Why? Note that the estimator $\hat{\sigma}^2$ is an unbiased and consistent estimator for σ^2 . Let's write first the matrix $P = X(X'X)^{-1}X'$ and $M = I - P$. M is symmetric and idempotent

$$MM = M^2 = M$$

and $MX = (I - X(X'X)^{-1}X')X = X - X = 0$. Note that

$$\begin{aligned}E(\hat{u}'\hat{u}) &= E((y - X\hat{\beta})'(y - X\hat{\beta})) = E((y - Py)'(y - Py)) \\ &= E((My)'(My)) = E((Mu)'(Mu)) = E(u'Mu) \\ &= E(\text{trace}(u'Mu)) = \text{trace}(E(u'u)M) = \text{trace}(E(uu')M) \\ &= \text{trace}(\sigma^2 IM) = \sigma^2 \text{trace}(M) \\ &= \sigma^2 \text{trace}(I - X(X'X)^{-1}X') = \sigma^2(n - p).\end{aligned}$$

Therefore, $E\hat{u}'\hat{u}/n - p = \sigma^2$.

It is also consistent,

$$\hat{\sigma}^2 = \frac{u'u}{n-p} - \frac{u'X(X'X)^{-1}X'u}{n-p}$$

The first term is, under A4,

$$\frac{\sum_{i=1}^n u_i^2}{n-p} \rightarrow \sigma^2.$$

The second term, under A3,

$$\frac{1}{n-p} \sum_{i=1}^n x_i' u_i \rightarrow 0$$

We know that $\frac{1}{n}(X'X) \rightarrow Q_0$, and then by continuity of inverse operation, we have that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n}(X'X) \right)^{-1} \rightarrow Q_0^{-1}.$$

By Slutsky, we have that,

$$\begin{aligned} \lim_{n \rightarrow \infty} \hat{\sigma}^2 &= \lim_{n \rightarrow \infty} \left(\frac{u'u}{n-p} - \frac{u'X(X'X)^{-1}X'u}{n-p} \right) \\ &= \sigma^2 \end{aligned}$$

The distribution of our test statistic is t -student with $n-r$ degrees of freedom. Note that,

$$\begin{aligned} \frac{R\hat{\beta} - \delta}{\sqrt{\hat{\sigma}^2 R(X'X)^{-1}R'}} &= (\hat{\sigma}^2 R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= ((\hat{u}'\hat{u}/(n-p))R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= ((\hat{u}'\hat{u}/(n-p)\sigma^2)\sigma^2 R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= (\hat{u}'\hat{u}/(n-p)\sigma^2)^{-1/2} (\sigma^2 R(X'X)^{-1}R')^{-1/2} (R\hat{\beta} - \delta) \\ &= (\hat{u}'\hat{u}/(n-p)\sigma^2)^{-1/2} \times \mathcal{N}(0, 1) \\ &= \frac{\mathcal{N}(0, 1)}{\sqrt{\frac{(n-r)\hat{\sigma}^2}{\sigma^2}/n-r}} \\ &= \frac{N(0, 1)}{\sqrt{\chi_{n-r}^2/n-r}}, \end{aligned}$$

because $\hat{\sigma}^2$ and $\hat{\beta}$ are independent.

Example 3. Suppose $R = (1, \dots, 0)$ and $\delta = 0$.

$$R\hat{\beta} - \delta = \hat{\beta}_1$$

and $R(X'X)^{-1}R' = (X'X)^{-1}_{ii}$. Therefore, we can test $H_0 : \beta_1 = 0$, by,

$$\frac{\hat{\beta}_1}{\hat{\sigma}\sqrt{(X'X)^{-1}_{ii}}} = \frac{\hat{\beta}}{se(\hat{\beta}_1)} \sim t_{n-1}.$$

where se denotes standard error. The null hypothesis $H_0\beta_j = 0$ can be tested by considering

$$\frac{\hat{\beta}_j}{se(\hat{\beta}_j)}.$$

Example 4. Consider the model,

$$wages_i = \beta_0 + \beta_1 educ_i + \beta_2 exp_i + \beta_3 tenure_i + u_i$$

Consider the example in Wooldridge's book.

$$\begin{array}{rcccc} \log(wage) & = & 0.284 & + & 0.092educ & + & 0.0041exp & + & 0.022tenure \\ & & (0.104) & & (0.007) & & (0.0017) & & (0.003) \end{array}$$

We would like to test if experience is a significant factor to explain wages. Consider:

$$H_0 : \beta_{exp} = 0; \quad H_a : \beta_{exp} > 0$$

Our t statistic is

$$\hat{t} = \frac{\hat{\beta}_{exp}}{se(\hat{\beta}_{exp})} = \frac{0.0041}{0.0017} = 2.4$$

Since the degrees of freedom is $n - p - 1 = 526 - 3 - 1 = 522$, we can use the standard normal critical values. Then $\hat{t} = 2.4 > 1.64 = c$ we reject the null hypothesis, and we conclude that experience is statistically significantly different from zero at 5 percent. We conclude also that $2\frac{1}{2}$ years of experience would increase wages by 1 percent.

Alternatively we consider the null and alternative hypotheses,

$$H_0 : \beta = 0; \quad H_a : \beta \neq 0$$

In this case, economic theory does not help to determine if the parameter should be positive or negative, so the alternative hypothesis reflects the lack of information. The rejection rule is,

$$|t| > c$$

Note that we are considering the absolute value of the t statistic. If we reject the null hypothesis, we say that x is statistically significantly different from zero. If we cannot reject the null hypothesis that $\beta = 0$, we say that x is not statistically significantly different from zero. We never say that we accept the null hypothesis.

INSERT FIGURE ONE and TWO-SIDED TEST

Example 5. Suppose you derive that the demand equation of a good is $Q = c/P$, where Q is quantity, P is price, and c is a positive constant. In this case, the demand elasticity is -1. We can formulate a regression version of the demand to test

$$\log(Q) = \log(c) + \beta \log(P) + u$$

The null hypothesis is $H_0 : \beta = -1$. Under the null hypothesis, the appropriate t statistic is

$$\hat{t} = \frac{\hat{\beta} + 1}{se(\hat{\beta})}$$

Remark 1. (P-values) It is defined as the smallest significance level at which the null hypothesis would be rejected. For instance, if the t statistic is $\hat{t} = 0.0039/0.0022 = 1.77$, we can compute the p-value as

$$\text{p-value} = P(T > 1.77) = 1 - 0.9616 = 0.0384$$

Therefore, we conclude that we reject the null hypothesis at 10 and 5 percent, but we cannot reject the null hypothesis at 1 percent significance level.

INSERT FIGURE

What if we have r independent restrictions? Consider first the following two results:

If $Z \sim N(M, \Omega)$ and $k = \text{rank}(\Omega)$, then

$$(Z - M)' \Omega^{-1} (Z - M) \sim \chi_k^2.$$

If χ_1^2 , has n_1 degrees of freedom and χ_2^2 has n_2 degrees of freedom, then,

$$\frac{\chi_1^2/n_1}{\chi_2^2/n_2} \sim F_{n_1, n_2}.$$

By result 1, because $R\hat{\beta} \sim N(R\beta, \sigma^2 R(X'X)^{-1}R')$, we have that,

$$(R\hat{\beta} - \delta)' \sigma^{-2} (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta) \sim \chi_r^2$$

The feasible version of the test statistic includes $\hat{\sigma}^2$, and therefore, it can be written as,

$$\begin{aligned} (R\hat{\beta} - \delta)' \frac{1}{\hat{\sigma}^2} [R(X'X)^{-1}R']^{-1} (R\hat{\beta} - \delta) / r &= \frac{(R\hat{\beta} - \delta)' \frac{1}{\hat{\sigma}^2} (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta) / r}{(u'Mu/\sigma^2)/n - p} \\ &= \frac{(R\hat{\beta} - \delta)' (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta)}{r\hat{\sigma}^2} \sim F_{r, n-p}. \end{aligned}$$

The test statistic is,

$$F = \frac{(R\hat{\beta} - \delta)' (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - \delta)}{r\hat{\sigma}^2} \quad (2)$$

Example 6. Consider that we want to impose q restrictions e.g. $\beta_1 = \beta_2 = \dots = \beta_q = 0$. In other words, we want to evaluate if the group of variables $x_1, x_2, \dots, x_q$ has no significant effect on the dependent variable. Our null hypothesis will be

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_q = 0 \quad H_a : \text{at least one } \beta_j \neq 0,$$

which suggests that under the null hypothesis the true model is

$$y = \beta_0 + \beta_{q+1}x_{q+1} + \dots + \beta_px_p + u \quad (3)$$

instead of the model

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_qx_q + \dots + \beta_px_p + u \quad (4)$$

The first model is the restricted model because we are imposing q restrictions. The second model is the unrestricted since we are not imposing a restriction. To evaluate the null hypothesis, we use the F test, which is defined as

$$F \equiv \frac{(SSR_r - SSR_u)/q}{SSR_u/(n - p - 1)}$$

where SSR_r is the sum of squared residuals from the restricted model and SSR_u is the sum of squared residuals from the unrestricted model. The F statistic and the R^2 of a regression has a close relation since both statistics depend on the sum of squared residuals. The derivation of the exact relation is left as an exercise.

Example 7. It is useful to evaluate the group of variables $x_1, x_2, \dots, x_q$ has no effect on the

dependent variable, which is basically an evaluation of the regression model. This is the case where none of the explanatory variables has a significant effect on the dependent variable,

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \quad H_a : \text{at least one } \beta_j \neq 0,$$

When we reject the null hypothesis using an F test, we say that the group of variables are jointly statistically significant at the appropriate significance level.

2 Performance of tests under non-Gaussian assumptions

The classical t and F statistic might be used under non-normality conditions. However, as it will be clear in the cases below, a necessary condition is that the sample should be reasonable large. Another important change is associated with the critical values. Instead of looking at the t distribution and F distribution for critical values, we consider the standard normal distribution for the t test and the chi-square distribution for the F test.

Assume that we are interested in the j -th covariate in a multivariate regression. We are interested in the hypothesis $\beta_j = 0$ is

$$t = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 (X'X)^{-1}_{jj}}} = \frac{\sqrt{n}\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 n (X'X)^{-1}_{jj}}}$$

where $(X'X)^{-1}_{jj}$ denotes the (j, j) -th element of the matrix $X'X^{-1}$. Under the null hypothesis that $\beta_{j0} = 0$, we have,

$$\sqrt{n}(\hat{\beta}_j) \rightarrow \mathcal{N}(0, \sigma_0^2 Q_{0,jj}^{-1}),$$

where Q_0 is the limiting form of the $X'X$. Then, because $\hat{\sigma}^2$ is a consistent estimator for the parameter σ_0^2 and $(X'X)/n$ is a consistent estimator for the unknown matrix Q_0 , we can apply the Slutsky theorem and obtain:

$$t = \frac{\sqrt{n}\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 n (X'X)^{-1}_{jj}}} \rightarrow \mathcal{N}(0, 1)$$

The argument for the F test is similar. Under the null hypothesis $R\beta_0 = r$, we have that

$$\sqrt{n}(R\hat{\beta} - r) = \sqrt{n}(R\hat{\beta} - R\beta_0) = \sqrt{n}R(\hat{\beta} - \beta_0) \rightarrow N(0, \sigma_0^2(RQ_0^{-1}R'))$$

Note also that $\hat{\sigma}^2 \rightarrow \sigma_0^2$ and

$$\frac{1}{n}[R(X'X)^{-1}R']^{-1} \rightarrow (RQ_0^{-1}R')^{-1}$$

Therefore

$$\hat{F} = \frac{(R\hat{\beta} - r)'[R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r)/(p - k)}{\hat{\sigma}^2} \rightarrow \chi_{p-k}^2$$

Heteroskedasticity and Clustered Standard Errors

Lecture 5

Recall that in a regression model,

$$y = x'\beta + u,$$

we assumed that $E(u|x) = 0$ and $V(u|x) = \sigma^2$. Heteroskedasticity occurs when $V(u_i|x) = \sigma_i^2$.

INSERT FIGURE HOMOCEDASTICITY-HETEROCEDASTICITY

If we use an estimator for $V(\hat{\beta})$ in the homocedastic case, we get a biased estimate of the true variance. Then, the classical inferential including the t and F tests, are invalidated since they use estimates of the variance of the estimator.

1 Generalized Least Squares

Suppose that $\mathbb{E}uu' = \Omega \neq \sigma^2 I_N$, and Ω is known. Consider a decomposition of the matrix $\Omega^{-1} = CC' = P\Lambda^{-1}P'$ where P is an orthogonal matrix and Λ is a diagonal matrix including the characteristic roots of Ω . It turns out that we can write

$$\mathbb{E}uu' = \Omega = (C')^{-1}C^{-1}.$$

We can then transform the regression as

$$\begin{aligned} C'y &= C'X\beta + C'u \\ \tilde{y} &= \tilde{X}\beta + \tilde{u} \end{aligned}$$

The error term vector $\tilde{u}$ is mean zero, and variance I (e.g. $\tilde{u} \sim \mathcal{N}(0, I)$) because the transformation achieves

$$V(\tilde{u}) = V(C'u) = C'V(u)C = C'(C')^{-1}C^{-1}C = I$$

Therefore, the generalized least squares estimator is

$$\hat{\beta} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y,$$

or equivalently,

$$\hat{\beta} = \arg \min ||y - X\beta||_{\Omega^{-1}}^2$$

The GLS estimator is unbiased with conditional variance equal to,

$$\begin{aligned} V(\hat{\beta}) &= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}V(y|X)\Omega^{-1}X(X'\Omega^{-1}X)^{-1} \\ &= (X'\Omega^{-1}X)^{-1}, \end{aligned}$$

which is “smaller” than the variance-covariance matrix of the OLS estimator in a model of non-spherical errors.

Theorem 1. $V(\hat{\beta}_{OLS}) - V(\hat{\beta}_{GLS})$ is a positive semi-definite matrix.

Proof: For simplicity C is symmetric. Note first that the conditional variance of the OLS is equal to

$$V(\hat{\beta}_{OLS}) = V((X'X)^{-1}X'u) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$$

Consider the decomposition,

$$\begin{aligned} X'\Omega^{-1}X &= (X'CC'X) = W'W \\ X'\Omega X &= (X'(C')^{-1}C^{-1}X) = S'S \\ X'X &= (X'CC^{-1}X) = W'S \end{aligned}$$

Then, we have that

$$(X'X)(X'\Omega X)^{-1}(X'X) = W'S(S'S)^{-1}S'W = W'P_SW \leq W'W$$

■

The GLS estimator is the weighted least squares (WLS) estimator when we only have heterocedasticity,

$$C^{-1} = \Omega^{1/2} = \text{diag}(\sigma_i)$$

Therefore we can re-weight by $\Omega^{-1/2}$ instead of C .

1.1 Estimation

The GLS estimator is unfeasible meaning that it can not be implemented in practice. The matrix Ω is typically unknown, thus we would like to implement a method to estimate it. We

can consider instead $\Omega(\gamma)$, and after establishing a relation between the conditional variance and the parameter γ , we may be able to implement the method.

Consider for simplicity

$$y_i = x_i\beta + (z_i'\gamma)^{1/2}u_i,$$

which implies that the i th diagonal element of Ω is $\Omega_{ii} = (z_i'\gamma)$. Note that if we estimate γ , we can estimate Ω . A simple estimation procedure considers:

1. Obtain the OLS estimator of γ by running,

$$(y_i - x_i\hat{\beta})^2 = z_i'\gamma + v_i$$

2. Plug $\hat{\gamma}$ on Ω , and obtain

$$\hat{\beta} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y.$$

Other alternative convenient first steps exists in the literature for the multiplicative, quadratic, etc. models of heterocedasticity. For instance, the conditional variance of u could be assumed to be $\exp(\gamma'x)$.

For completeness, we review the consequences of ignoring heterocedasticity:

- The OLS estimator is unbiased but not efficient.
- The estimate of the variance of the OLS estimator is a biased estimator of the true variance. The classical testing procedures are invalidated.

2 Weighted Least Squares: A simple example

Consider the simple regression model without intercept,

$$y_i = \beta x_i + u_i \tag{1}$$

where the conditional variance is

$$V(u_i|x_i) = \sigma_i^2.$$

For simplicity we assume that

$$V(u_i|x_i) = \sigma_i^2 = \sigma^2 x_i$$

We use the simple model presented before to illustrate the effects of heteroscedasticity. The OLS estimator of β is,

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

The variance of the OLS estimator is in this case,

$$V(\hat{\beta}) = \frac{\sum_i^n x_i^2 \sigma_i^2}{(\sum_{i=1}^n x_i^2)^2}$$

We now consider an estimator that weighs each observation by $1/\sqrt{x_i}$,

$$\frac{y_i}{\sqrt{x_i}} = \beta \frac{x_i}{\sqrt{x_i}} + \frac{u_i}{\sqrt{x_i}} \Rightarrow y_i^* = \beta x_i^* + u_i^* \quad (2)$$

because $V(u_i/\sqrt{x_i}|x_i) = x_i^{-1}V(u_i|x_i) = x_i^{-1}\sigma^2 x_i = \sigma^2$. Consequently, the estimator of the slope parameter is,

$$\tilde{\beta} = \frac{\sum_{i=1}^n x_i^* y_i^*}{\sum_{i=1}^n x_i^{*2}}$$

and the variance of $\tilde{\beta}$ is,

$$V(\tilde{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n x_i^{*2}} = \frac{\sigma^2}{\sum_{i=1}^n (x_i/\sqrt{x_i})^2}$$

Now we can compare the variance of the two estimators to see if, under heteroscedasticity, the OLS estimator is BLUE (given that it is unbiased we only have to check if it is minimum variance). Taking the ratio of the two estimators,

$$\frac{\text{Var}(\hat{\beta})}{\text{Var}(\tilde{\beta})} = \frac{\sum_{i=1}^n x_i^3 \sum_{i=1}^n x_i}{(\sum_{i=1}^n x_i^2)^2},$$

Recall that the squared correlation coefficient between a and b is

$$r^2 = \frac{(\sum ab)^2}{(\sum a^2)(\sum b^2)} \leq 1 \Rightarrow (\sum ab)^2 \leq (\sum a^2)(\sum b^2)$$

The last expression is a version of the Cauchy-Schwarz Inequality. Note that in our case, $a^2 = x^3$ and $b^2 = x$. Then,

$$\frac{\text{Var}(\hat{\beta})}{\text{Var}(\tilde{\beta})} \geq 1,$$

which implies that the OLS estimator is inefficient under heteroscedasticity.

3 Testing Heteroscedasticity:

We start assuming a multivariate regression model,

$$y_i = x_i' \beta + u_i$$

We assume that the null hypothesis is

$$H_0 : V(u|x) = \sigma^2,$$

that is, our null hypothesis is no heteroscedasticity. Since $E(u|x) = 0$, we write H_0 as

$$H_0 : E(u^2|x) = E(u^2) = \sigma^2,$$

Note that we need to evaluate the conditional (on the independent variables) expectation of the squared of the error term,

$$u^2 = \alpha_0 + \alpha_1 x_1 + \dots + \alpha_p x_p + \varepsilon,$$

The homocedastic case implies that $\alpha_1 = \alpha_2 = \dots = \alpha_p = 0$, therefore

$$E(u^2|x_1, \dots, x_p) = \alpha_0$$

Note however this is unfeasible because we do not know the error term u^2 . A simple approach is to replace it by $\hat{u}^2$ and estimate,

$$\hat{u}_i^2 = \alpha_0 + \alpha_1 x_{1i} + \dots + \alpha_p x_{pi} + \varepsilon_i$$

and test if the slope coefficients are all equal to zero,

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_p = 0; \quad H_a : \text{At least one } \alpha_j \neq 0$$

using a F test. If $F > c$, where c is determined by a value of statistical significance, we reject the null hypothesis of no heterocedasticity.

The Lagrange multiplier (LM) version the test is very simple to compute since

$$LM = n \times R_{\hat{u}^2}^2,$$

which is distributed as χ_p^2 under the null hypothesis of homocedasticity. This is the Breusch-Pagan (1980) test for heteroscedasticity.

A variation of the test was developed by White (1980). To simplify the presentation, assume that $p = 2$:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i,$$

The White's test is based on the estimation of

$$u_i^2 = \alpha_0 + \alpha_1 x_{1i} + \alpha_2 x_{2i} + \alpha_3 x_{1i}^2 + \alpha_4 x_{2i}^2 + \alpha_5 x_{1i} x_{2i} + \varepsilon_i,$$

The null hypothesis in this case is,

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_5 = 0; \quad H_a : \text{at least one } \alpha_j \neq 0,$$

which can be evaluated with a *LM* or *F* test statistic.

4 A heteroscedasticity-robust matrix

The covariance matrix of the error term, under homocedasticity, is equal to,

$$\Sigma = E(uu') = E(uu'|x) = E(\text{diag}(u_1^2, u_2^2, \dots, u_n^2)|x)$$

or alternatively,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{bmatrix}$$

Therefore, under heteroskedasticity, the variance of the estimator is,

$$\begin{aligned} V(\hat{\beta}|X) &= V((X'X)^{-1}X'y) \\ &= (X'X)^{-1}X'V(y|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'V(u|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\Sigma X(X'X)^{-1} \end{aligned}$$

If we know the error term, then notice that $(uu')_{ii} = u_i^2$ and the conditional expected value of the variance is,

$$\begin{aligned} E(V(\hat{\beta}|X)) &= (X'X)^{-1}X'E(\text{diag}(u_1^2, u_2^2, \dots, u_n^2|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\Sigma X(X'X)^{-1} \end{aligned}$$

In practice, we obtain least square residuals, $\hat{u}_i^2 = (y_i - x_i'\hat{\beta})^2$, where $\hat{\beta}$ is the least square

estimator of β . We can then form a matrix,

$$\hat{\Sigma} = \begin{bmatrix} \hat{u}_1^2 & 0 & \dots & 0 \\ 0 & \hat{u}_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \hat{u}_n^2 \end{bmatrix},$$

and construct a feasible estimator:

$$\hat{V}(\hat{\beta}|X) = (X'X)^{-1}X'\hat{\Sigma}X(X'X)^{-1}.$$

This estimator is called ‘heteroscedasticity-robust’ estimator of the covariance matrix. It was introduced to the econometrics literature in an influential paper by White (1980). There are other alternatives typically implemented in statistical packages for $\hat{u}^2$ (e.g., Andrews 1991). To simplify this, if $p = 1$, White (1980) suggests to obtain the standard error of the OLS estimator

$$se(\hat{\beta}_1) = \left(\frac{\sum_{i=1}^n (x_i - \bar{x})^2 \hat{u}_i^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \right)^{1/2}.$$

Note that the estimator represents a consistent estimator of the unknown covariance. In small samples, unfortunately, the estimator might be biased. To see this, recall that $\hat{u}\hat{u}' = uMu' = yMy'$. Then,

$$\begin{aligned} \hat{u}_i^2 &= (y_i - x_i'\hat{\beta})^2 = (y_i + x_i'\beta - x_i'\beta - x_i'\hat{\beta})^2 \\ &= (u_i - x_i'(\hat{\beta} - \beta))^2 \\ &= (u_i - x_i'(X'X)^{-1}X'u)^2 \\ &= u_i^2 - 2u_ix_i'(X'X)^{-1}X'u + x_i'(X'X)^{-1}X'uu'X(X'X)^{-1}x_i \end{aligned}$$

Under homocedasticity, and denoting the so-called leverage point $h_{ii} = x_i'(X'X)^{-1}x_i$, we have that,

$$\begin{aligned} E(\hat{u}_i^2) &= E(u_i^2 - 2u_ix_i'(X'X)^{-1}X'u + x_i'(X'X)^{-1}X'uu'X(X'X)^{-1}x_i) \\ &= E(u_i^2) - 2E(u_ix_i'(X'X)^{-1}X'u) + E(x_i'(X'X)^{-1}X'uu'X(X'X)^{-1}x_i) \\ &= \sigma^2 - 2\sigma^2h_{ii} + \sigma^2h_{ii} \\ &= \sigma^2(1 - h_{ii}) < \sigma^2 = E(u^2) \end{aligned}$$

Then, insisting with the simple example when $p = 1$, we have that,

$$\begin{aligned} E(\hat{V}(\hat{\beta}_1)) &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2 E(\hat{u}_i^2)}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma^2 (1 - h_{ii})}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \\ &= \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} - \frac{\sum_{i=1}^n (x_i - \bar{x})^2 h_{ii} \sigma^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \end{aligned}$$

indicating that $\hat{V}(\hat{\beta}|X)$ is downward bias (note that the first term is the variance of the least squares estimator under homocedasticity). An standard correction is to divide $\hat{u}_i^2$ by $1 - h_{ii}$.

5 Consistent Estimation of S

Define,

$$\hat{S} := \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 x_i x_i',$$

where $\hat{u} = y - x_i' \hat{\beta}$, and $\hat{\beta}$ is a consistent estimator for the parameter β .

To show the consistency of $\hat{S}$ for $S = E(x_i x_i' u_i^2)$ we need a condition on the fourth moment of x_i as stated before in Assumption A5.

Theorem 2. (*Consistency of $\hat{S}$*) Suppose S exists and it is finite and $\hat{\beta}$ is a consistent estimator for β . Then, $\hat{S} \rightarrow S$.

The formal proof will be discussed later, but I would like to provide a brief illustration considering $p = 1$. By definition,

$$\begin{aligned} \hat{u}_i^2 &= (y - x_i \hat{\beta})^2 = (y - x_i \beta + x_i \beta - x_i \hat{\beta})^2 = (u_i - x_i(\hat{\beta} - \beta))^2 \\ &= u_i^2 - 2u_i x_i(\hat{\beta} - \beta) + (\hat{\beta} - \beta)^2 x_i^2, \end{aligned}$$

and therefore,

$$\hat{u}_i^2 - u_i^2 = -2u_i x_i(\hat{\beta} - \beta) + (\hat{\beta} - \beta)^2 x_i^2.$$

It follows that,

$$\begin{aligned}
 \hat{S}_n - S_n &= \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 x_i^2 - \frac{1}{n} \sum_{i=1}^n u_i^2 x_i^2 \\
 &= -2(\hat{\beta} - \beta) \frac{1}{n} \sum_{i=1}^n u_i^2 x_i^3 + (\hat{\beta} - \beta)^2 \frac{1}{n} \sum_{i=1}^n (x_i^2)(x_i^2) \\
 &= -2(\hat{\beta} - \beta) \frac{1}{n} \sum_{i=1}^n u_i^2 x_i^3 + (\hat{\beta} - \beta)^2 \frac{1}{n} \sum_{i=1}^n x_i^4.
 \end{aligned}$$

Therefore, under the existing assumptions,

$$\hat{S}_n = \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 x_i^2 \rightarrow E(u_i x_i^2) =: S,$$

because the moments are bounded and the estimator $\hat{\beta}$ is consistent for β under Assumptions A1-A5.

6 Clustered Standard Errors

This section presents a few ideas on the estimation of standard errors in the presence of “group” effects. Most of the corrections you’ll find below are available in statistical packages, but I thought it might be a good idea to have a lecture discussing specifically potential issues in empirical research.

Suppose we have

$$y_i = x_i' \beta + u_i \tag{3}$$

and we use OLS to estimate the unknown parameter β . It is known that, if $Var(u_i) = \sigma^2$,

$$\hat{\beta} = (X'X)^{-1} X'y \tag{4}$$

$$Var(\hat{\beta}) = \sigma^2 (X'X)^{-1}. \tag{5}$$

If $Var(u_i) = \sigma_i^2$, we have that,

$$Var(\hat{\beta}) = (X'X)^{-1} X' \Omega X (X'X)^{-1} \tag{6}$$

where $\Omega = E u u'$. In terms of estimation, it is known that the unknown parameters can be estimated by

$$\hat{\sigma}^2 = \left(\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 \right), \tag{7}$$

$$\hat{\Omega} = \text{diag}(\hat{u}_i^2), \quad (8)$$

where $\hat{u}_i = y_i - x_i' \hat{\beta}$ is a residual. Note that $\hat{\Omega} \rightarrow \Omega$ if $\hat{\beta} \rightarrow \beta$. Therefore, the robust standard errors relies in an asymptotic approximation and may be biased in small samples.

Suppose now that we have $\{(y_i, x_i)\}_{i=1}^n$, but we know that the model may vary by clusters $c = 1, \dots, C$:

$$y_{ic} = x_{ic}' \beta_c + u_{ic}. \quad (9)$$

Although it is reasonable to think that $\text{Cov}(u_{ic}, u_{is}) = 0$, it may be the case that

$$\text{Cov}(u_{ic}, u_{jc}) \neq 0. \quad (10)$$

We start assuming that the error term u_{ic} is not independent of x_{ic} . For instance, if $u_{ic} = \alpha_c + v_{ic}$, then $\text{Cov}(x_{ic}, \alpha_c) \neq 0$. In this case, clusters introduce biases if the model does not condition on α_c . Therefore, one might be interested in estimating,

$$E(y_{ic}|X_{ic}, \alpha_c) = x_{ic}' \beta + \alpha_c, \quad (11)$$

when the number of clusters are large. The model can be estimated by classical least-squares transformations.

In the case of random effects, we can consider stacking observations within clusters,

$$y_c = X_c \beta + u_c \quad (12)$$

where y_c is a vector of dimension $n_c \times 1$, and X_c is a $n_c \times p$ matrix. Let $n = \sum_c n_c$. Then,

$$\hat{\beta} = (X'X)^{-1} X'y = \left(\sum_{c=1}^C \sum_{i=1}^{n_c} x_{ic} x_{ic}' \right)^{-1} \sum_{c=1}^C \sum_{i=1}^{n_c} x_{ic} y_{ic}.$$

Notice that $\text{Var}(\hat{\beta}) \neq \sigma^2 (X'X)^{-1}$. The clustered-robust standard error is,

$$(X'X)^{-1} X' \hat{\Omega} X (X'X)^{-1} = \left(\sum_{c=1}^C X_c' X_c \right)^{-1} \left(\sum_{c=1}^C X_c' \hat{u}_c \hat{u}_c' X_c \right) \left(\sum_{c=1}^C X_c' X_c \right)^{-1} \quad (14)$$

where the residuals are $\hat{u}_c = y_c - X_c \hat{\beta}$. The problem with this correction is that we need $C \rightarrow \infty$ for consistency.

If we assume that $u_{ic} = \alpha_c + v_{ic}$, we can adopt the random effect framework by assuming that α and v are independent. Notice that

$$E(u_{ic} u_{jc}) = E(\alpha_c^2 + v_{ic} \alpha_c + v_{jc} \alpha_c + v_{ic} v_{jc}) \quad (15)$$

and then, if $i = j$,

$$E(u_{ic}u_{ij}) = \sigma_\alpha^2 + \sigma_v^2 = \sigma^2, \quad (16)$$

and, if $i \neq j$,

$$E(u_{ic}u_{jc}) = \sigma_\alpha^2 = \rho\sigma^2, \quad (17)$$

where $\rho = \sigma_\alpha^2/(\sigma_u^2 + \sigma_\alpha^2)$. By results similar to the Balesta - Nerlove estimator, we have that

$$\Sigma_c = E(u_c u_c') = \sigma^2((1 - \rho)I_c + \rho 1_c' 1_c) \quad (18)$$

It then follows that the variance of the $\hat{\beta}$ *OLS* is,

$$Var(\hat{\beta}) = \left(\sum_{c=1}^C X_c' X_c\right)^{-1} \sum_{c=1}^C \sigma^2 X_c' ((1 - \rho)I_c + \rho 1_c' 1_c) X_c \left(\sum_{c=1}^C X_c' X_c\right)^{-1}. \quad (19)$$

A particular useful case to fix ideas is a model with individual invariant variables. Suppose now that the model is,

$$y_{ig} = \beta_0 + \beta_1 x_g + u_{ig}, \quad (20)$$

It is possible to show that,

$$V(\hat{\beta}_1) = (1 + (M - 1)\rho)V(\tilde{\beta}_1) \quad (21)$$

where $V(\tilde{\beta}_1)$ is the variance of standard *OLS* estimator. Note that the bias of the *OLS* could be severe.

It is possible to test for cluster effects following Moulton (1997). The null hypothesis is

$$H_0 : \rho = \frac{\sigma_\alpha^2}{(\sigma_u^2 + \sigma_\alpha^2)} = 0 \quad (22)$$

$$H_a : \rho > 0. \quad (23)$$

A one-sided *LM* test can be formulated as follows:

$$LM = \frac{\sum_c (N_c \bar{u}_c)^2 - \sum_c \sum_i \hat{u}_{ic}^2}{\hat{\sigma}^2 [2(\sum_c N_c^2 - N)]^{1/2}} \quad (24)$$

where $\hat{\sigma}^2 = \sum_c \sum_i \hat{u}_{ic}^2 / N$, where $\hat{u}_{ic}$ denotes the least-square residual and $\bar{u}_c$ is the average residual for cluster C .

Example 1. Moulton (1986) wage regressions considering CPS data. The number of individuals in the sample was $n \approx 18,000$ in $c = 49$ states. Although the intraclass correlation coefficient estimate $\hat{\rho}$ was 0.03, the bias was large because the clusters were large

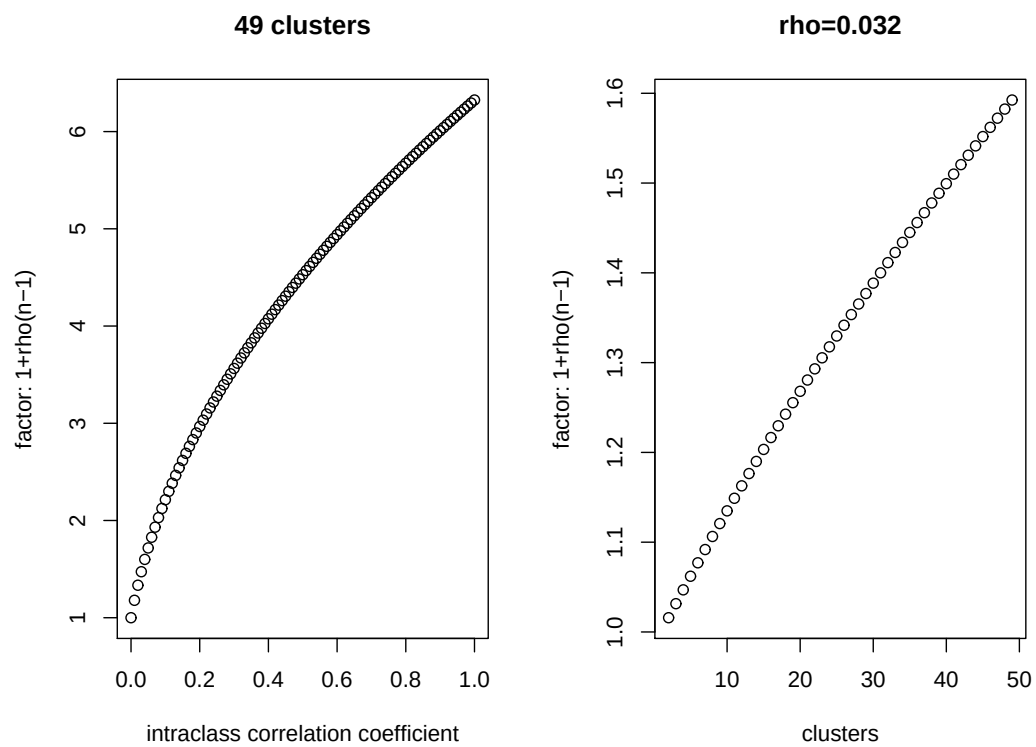


Figure 1: Standard Errors and the Moulton factor

($M \approx 380$). The Mounon factor, which is $((1 + (M - 1)\rho))^{.5}$, was 3.7 leading to sizable biases in the standard errors.

Models with Endogenous Regressors

Lecture 6

1 Basics

Suppose that the true model is,

$$y = X\beta + u \tag{1}$$

If $\mathbb{E}(u|X) = 0$, it can be shown that the OLS estimator,

$$\hat{\beta} = \beta + (X'X)^{-1}X'u$$

is unbiased and consistent. Consistency can be shown by proving that the second term converges to zero in probability. First write the second term as

$$(n^{-1}X'X)^{-1}(n^{-1}X'u),$$

and then apply Slutsky theorem, to find that $\hat{\beta} - \beta \rightarrow 0$. However, sometimes even in large samples, the estimator may not be “close” (in some sense) to the true parameter. In the presence of endogenous independent variables (e.g. $\mathbb{E}(u|X) \neq 0$), the estimator $\hat{\beta}$ is biased and inconsistent,

$$\hat{\beta} \rightarrow \beta + Q_{xx}^{-1}Q_{xu} \neq \beta$$

where the matrices Q are limiting forms of $X'X$ and $X'u$. Note that the lack of orthogonality between the error term and the independent variables makes the OLS inconsistent.

1.0.1 Example: Errors in Variables

Many variables (GDP, permanent income, capital depreciation, intelligence, etc.) are measured with errors. Consider the model,

$$y_i = \beta'x_i^* + \varepsilon_i$$

where x_i^* is not observed and independent of ϵ_i . The variable that is observed by the researcher is x_i , which can be expressed as

$$x_i = x_i^* + u_i$$

The OLS estimate of the model,

$$y_i = x_i^{*'}\beta + \varepsilon_i = x_i'\beta + (\varepsilon_i - u_i'\beta) = x_i'\beta + v_i,$$

may be biased and inconsistent because $x \not\perp v$,

$$\mathbb{E}(x_i v_i) = \mathbb{E}(x_i^* + u_i)(\varepsilon_i - u_i'\beta) = -\mathbb{E}u_i u_i'\beta \neq 0$$

Note that

$$\begin{aligned}\mathbb{E}(\hat{\beta}|X) &= (\sum x_i x_i')^{-1} \sum x_i \mathbb{E}(y_i|x_i) \\ &= \beta + (\sum x_i x_i')^{-1} \sum x_i \mathbb{E}(v_i|x_i) \\ &= \beta + \text{bias}\end{aligned}$$

Moreover, in large samples,

$$\begin{aligned}\hat{\beta} &= \beta + (n^{-1} \sum x_i x_i')^{-1} (n^{-1} \sum x_i v_i) \\ &= \beta + (n^{-1} \sum x_i x_i')^{-1} (n^{-1} \sum (x_i^* + u_i)(\varepsilon_i - u_i'\beta)) \\ &= \beta + (n^{-1} \sum x_i x_i')^{-1} (n^{-1} \sum (x_i^* \varepsilon_i - x_i^* u_i' \beta + u_i \varepsilon_i - u_i u_i' \beta))\end{aligned}$$

Since the disturbances ε_i and u_i , and the unobserved variable x_i^* are independent,

$$\hat{\beta} \rightarrow \beta - (Q_{x^*} + Q_u)^{-1} Q_u \beta \neq \beta$$

where $Q_{x^*} = \lim_{n \rightarrow \infty} \sum x_i^* x_i^{*'}$ and $Q_u = \lim_{n \rightarrow \infty} \sum u_i u_i'$. First, note that if one independent variable is measured with error, then, in general, all the estimates are biased and inconsistent (Why?). Second, if $p = 1$ (e.g., one independent variable),

$$\hat{\beta} \rightarrow \frac{\sigma_{x^*}^2}{(\sigma_{x^*}^2 + \sigma_u^2)} \beta$$

suggesting that the greater the measurement error, the worse the bias.

1.0.2 Example: Omitted Variables

The omitting variable bias is created when we do not introduce an unobserved variable. For example, since ability is a latent variable, we cannot estimate the true model returns to education model. On the other hand, there are variables related with the unobserved variable called proxies. For example, the IQ score is a proxy for ability (e.g., the higher person's IQ, the higher the level of ability).

Now, the question is what are the consequences on estimation of using a proxy variable? We consider the return to education model to illustrate and discuss the main lessons.

$$\log(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 exp_i + \beta_3 abil_i + u_i \quad (2)$$

$$abil_i = \alpha_0 + \alpha_3 IQ_i + \varepsilon_i \quad (3)$$

Replacing equation (6) in (5), we have the following model

$$\log(wage_i) = \delta_0 + \beta_1 educ_i + \beta_2 exp_i + \delta_3 IQ_i + v_i,$$

where $\delta_0 = \beta_0 + \alpha_0\beta_3$, $\delta_3 = \alpha_3\beta_3$, and $v_i = \beta_3\varepsilon_i + u_i$.

By least squares methods, we can obtain unbiased estimates of β_1 , and β_2 since the error term and the independent variables $educ$ and exp are independent. Also, the OLS estimators are consistent, that is, converges in probability to the true parameter value as the sample size increases. However, this conclusion can be drastically altered. Suppose that instead of equation (3), the true model assumes that ability is also related with education e.g.

$$abil_i = \alpha_0 + \alpha_1 educ_i + \alpha_3 IQ_i + \varepsilon_i \quad (4)$$

Now, the model is,

$$\log(wage_i) = \delta_0 + \delta_1 educ_i + \beta_2 exp_i + \delta_3 IQ_i + v_i,$$

where δ_0 , δ_3 , v_i are defined as before, and $\delta_1 = \beta_1 + \alpha_1\beta_3$.

It can be shown that, as the sample size increases, the estimator of the return to education will tend to

$$\hat{\delta}_1 \rightarrow \beta_1 + \alpha_1\beta_3 \neq \beta_1,$$

which is different than the true parameter β_1 since $\alpha_1 \neq 0$. Thus, although we introduce a variable to account for the effect of the unobserved ability, we have an inconsistent estimator of the return to education parameter. For example, if years of education is positively related with ability ($\alpha_1 > 0$), and ability is positively related with earnings ($\beta_3 > 0$), the asymptotic bias is positive.

2 IV Estimation

We can consider replacing x with another variable, say z . The new variable, called instrumental variable, has to satisfy certain conditions¹:

1. The instrumental variable z has to be uncorrelated with the error term u , $E(zu) = 0$
2. The instrumental variable z has to be correlated with the endogenous variable x , $E(zx) \neq 0$

Consider the following linear regression model,

$$y = X_1\beta_1 + X_2\beta_2 + u = X\beta + u$$

where X_1 are exogenous variables (e.g. $E(u|X_1) = 0$) and X_2 are endogenous -observed- variables (e.g. $E(u|X_2) \neq 0$). Suppose also that there is a matrix of instruments Z . The idea is as follows: multiply each side of the equation by the matrix of instrumental variables Z ,

$$Z'y = Z'X\beta + Z'u$$

where Z may include X_1 as instrument for X_1 , and Z_2 as instrument for X_2 . Dividing by n and applying the Law of Large Numbers on $n^{-1}Z'u$, we get an expression for β that leads to the IV estimator. (Why?). The instrumental variable estimator is,

$$\hat{\beta}_{IV} = (Z'X)^{-1}Z'y$$

Under spherical errors, the variance of the estimator

$$V(\hat{\beta}_{IV}) = \hat{\sigma}_u^2(Z'X)^{-1}Z'Z(Z'X)^{-1}$$

Note that the IV estimator $\hat{\beta}_{IV}$ can be also obtained from the moment condition

$$Z'(y - X\beta) = 0,$$

which involve p equations and p unknowns.

The IV estimator is unbiased under assumptions 1 and 2.(Why?). Note that

$$E(\hat{\beta}_{IV}) = \beta,$$

¹Why the IV has to satisfy these two conditions? The first condition may be intuitive. The first condition requires that z and u are uncorrelated implying that if we are going to use another variable, at least, it has to fix the original problem, $\text{Cov}(x, u) \neq 0$. Could you explain the second condition?

because $E(Z'u) = 0$. Also it is consistent

$$\hat{\beta}_{IV} - \beta = (n^{-1}Z'X)^{-1}(n^{-1}Z'u) \rightarrow 0$$

provided that the limiting form of $Z'X$ is deterministic and nonsingular. (Conditions needed for asymptotic identification). The errors are asymptotically uncorrelated with the instruments, which follows from condition 1 by the law of large numbers. Note that this condition may hold even if condition 1 does not.

What are the instruments Z ? First since $X_1 \perp u$, X_1 can be (trivially) used as IV for itself. We then need additional instruments satisfying 1 and 2 to identify the model. The vector Z_2 are IV variables for X_2 , often denoted as excluded (from the model) exogenous variables.

The IV estimator has the following asymptotic representation,

$$\sqrt{n}(\hat{\beta}_{IV} - \beta) \rightsquigarrow \mathcal{N}(0, \sigma^2 Q_{zx}^{-1} Q_{zz} Q_{zx}^{-1})$$

where $Q_{zx} = \lim_{n \rightarrow \infty} n^{-1}Z'X$. If $Z = X$, the IV estimator is trivially the OLS estimator. Lastly, note that the relationship between Z and X in large samples place a key role for determining the asymptotic variance.

2.0.1 Example: IV Wald Estimator

Consider simple regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where $\mathbb{E}(u_i|x_i) \neq 0$. Our model's assumptions are (a) the error term has expected value zero, and (b) the correlation between the instrumental variable and the error term is zero. These population conditions impose condition over the data counterpart that can be stated as

$$E(u) = 0 \Rightarrow \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (5)$$

$$E(zu) = 0 \Rightarrow \sum_{i=1}^n (z_i(y_i - \beta_0 - \beta_1 x_i)) = 0 \quad (6)$$

From the first equation, we obtain $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Replacing this expression in the second equation gives the Wald estimator,

$$\sum_{i=1}^n z_i(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^n z_i(x_i - \bar{x})$$

The variable $z_i = \{0, 1\}$ is a dummy variable, therefore we can write,

$$\sum z_i(y_i - \bar{y}) = N_1(\bar{y}_1 - \bar{y}) = N_1\left(\frac{N_0\bar{y}_1 + N_1\bar{y}_1}{N} - \frac{N_0\bar{y}_0 + N_1\bar{y}_1}{N}\right) = \frac{N_1N_0}{N}(\bar{y}_1 - \bar{y}_0),$$

Consequently, we can write the estimator as,

$$\hat{\beta}_{Wald} = \frac{(\bar{y}_1 - \bar{y}_0)}{(\bar{x}_1 - \bar{x}_0)}$$

3 Generalized Instrumental Variable

Note that $(Z'X)$ invertible needs to have a Z matrix with p instruments. When the number of IVs is equal to the number of parameters to be estimated, we say that the model is exactly identified. Often the number of IVs is greater than the number of parameters, so we can consider the following matrix

$$W = P_z X$$

to achieve a transformation. The model is said to be overidentified, and in this case the moment condition is

$$W'(y - X\beta) = 0,$$

giving the generalized instrumental variable estimator,

$$\hat{\beta} = (W'X)^{-1}W'y$$

The generalized IV estimator is usually known as the Two Stage Least Square (2SLS) estimator,

$$\hat{\beta}_{2SLS} = (X'P_zX)^{-1}X'P_zy$$

which is based on two projections²:

1. Regress X on Z to obtain,

$$\hat{X} = Z(Z'Z)^{-1}Z'X = P_zX$$

²The main idea of the 2SLS estimator is the following. In a binary regression model consider that the independent variable is endogenous:

$$y_i = x_i'\beta + u_i; \quad x_i = z_i'\pi + v_i$$

The second equation is interesting since it contains two parts that provides the decomposition needed to solve the endogeneity in the main equation $\pi_0 + \pi_1 z_i$ is uncorrelated with u but v_i is correlated with u . The idea of 2SLS is to use only the first part, the part of the decomposition that it is independent of the error term. The problem is that $z_i'\pi$ is not observed, but it can be predicted using the simple OLS regression $\hat{x}_i = z_i'\hat{\pi}$.

It is important to see the components of the projection matrix $\hat{X} = [P_z X_1, P_z X_2]$. Recall that $Z = [X_1, Z_2]$, therefore,

$$\hat{X} = [X_1, \hat{X}_2]$$

2. Then regress y on $\hat{X}$, obtaining the 2SLS estimator $\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1}\hat{X}'y$.

If the errors are homocedastic, the previous conditions (1)-(2) are satisfied, and the rank condition (as many IV's as independent variables) is satisfied, we have the following result.

Theorem 1. *The 2SLS estimator is asymptotically normally distributed with mean zero and variance matrix,*

$$\text{plim}_{n \rightarrow \infty} [n^{-1} \sigma^2 (X' P_z X)]^{-1}$$

Proof: Write the IV estimator as,

$$\hat{\beta} = \beta + (X' P_z X)^{-1} X' P_z u$$

Then, multiply each term by $\sqrt{n}$ and write it as,

$$\sqrt{n}(\hat{\beta} - \beta) = (n^{-1} X' P_z X)^{-1} (n^{-1/2} X' P_z u) + o_p(1)$$

The result follows passing n to ∞ and applying Slutsky theorem. ■ (Can you modify the result for the case of non-spherical errors?).

3.1 A Third Derivation

An interesting interpretation of the 2SLS is as follows. Define the residuals of X on Z ,

$$R = X - \hat{X} = X - P_z X = M_z X$$

Consider now the least square estimator β from a model that includes the residuals,

$$y = X\beta + R\gamma + u,$$

which is equal to

$$\hat{\beta}_{cv} = (X' M_R X)^{-1} X' M_R y$$

where cv stands for control variate. This estimator is identical to the 2SLS estimator because,

$$\begin{aligned} X' M_R &= X'(I - R(R'R)^{-1}R') = X'(I - (X - P_z X)(X' M_z X)^{-1}R') \\ &= X' - R' = X' - (X - P_z X)' = X' P_z \end{aligned}$$

Thus the “endogeneity bias” can be viewed as an omitted variable bias.

4 Tests

4.1 A misspecification test

Of course, in practical situations, we do not know whether the independent variables are correlated with the error term. We may use economic theory to guide us, but the real statistical question is simply: Is the endogeneity issue serious enough? For example, if the answer is no, we may ignore the issue and simply estimate the model using OLS.

The previous question can be examined with a test proposed by J. Hausman in 1978. The test about specification models in econometrics has a wide applicability to different models. In panel data, as it will be clear in future lectures, it helps to decide between the fixed and random effects models by testing the assumed hypothesis of independence between the individual specific effects and the independent variables.

In our cross-sectional setting, the null and alternative hypothesis may be formulated as,

$$H_0 : \mathbb{E}(X'u) = 0; \quad H_a : \mathbb{E}(X'u) \neq 0,$$

Under H_0 the OLS and IV estimators are consistent, though the OLS is efficient. Under H_a , the IV estimator is still consistent, but the OLS is inconsistent.

Therefore, a test based on the difference between these two estimators may indicate if the endogeneity affect the consistency of the OLS estimator. The idea is to evaluate whether the distance between the two estimators is asymptotically zero,

$$\hat{q} = \tilde{\beta} - \hat{\beta} \rightarrow 0$$

where $\tilde{\beta}$ is the IV estimator, $\hat{\beta}$ is the OLS estimator. What is the distribution of q ? We expect that q is asymptotically normal with asymptotic variance,

$$V(\tilde{\beta}) - V(\hat{\beta})$$

Lemma 1. Under H_0 , $V(\tilde{\beta} - \hat{\beta}) = V(\tilde{\beta}) - V(\hat{\beta})$

Proof: Write

$$\Delta = \tilde{\beta} - \hat{\beta} = [(X'P_zX)^{-1}X'P_z - (X'X)^{-1}X']u = [A - B]u$$

Then,

$$\mathbb{E}\Delta\Delta' = \mathbb{E}[(A - B)uu'(A' - B')] = \sigma_u^2[AA' - AB' - BA' + BB']$$

Note that

$$AB' = (X'P_zX)^{-1}X'P_zX(X'X)^{-1} = (X'X)^{-1} = BA' = BB',$$

so

$$\mathbb{E}\Delta\Delta' = \sigma^2[AA' - BB']$$

■

A natural test statistics is then,

$$h = \hat{q}'(V(\tilde{\beta}) - V(\hat{\beta}))^{-1}\hat{q}$$

which is asymptotically distributed as χ^2 . If h is low, say less than χ^2 critical value, we do not reject the null hypothesis. We conclude that the endogeneity does not affect the consistency of the OLS estimator. If h is high, say greater than χ^2 critical value, we reject the null hypothesis, and we conclude that endogeneity is a problem for the OLS estimator. Thus, we should use the IV estimator.

The Hausman specification test can be also used to evaluate two 2SLS estimators just by testing the exogeneity of the dubious IV variable. We can maintain the exogenous set of instruments and then we ask whether introducing the dubious IV variable is a good or bad idea.

4.2 Overidentifying Restrictions

Consider the regression model is well specified, $y = X\beta + u$. A candidate Z must satisfy,

$$E(u|Z) = 0 \text{ or } E(Z'u) = 0$$

Note that in the case of just indentified models, we have that

$$Z'\hat{u} = 0,$$

so we cannot test the previous conditions. However, if we have more instruments (say k) than endogenous variables (say q), we can consider the following test,

$$T = \hat{\sigma}^{-2}\hat{u}'Z(Z'Z)^{-1}Z'\hat{u} = \hat{\sigma}^{-2}\hat{u}'P_Z\hat{u} \sim \chi^2_{k-q},$$

under the null hypothesis. If we reject the null, this should be interpreted as evidence that instruments are endogenous. Let me present you a simple way to implement the test:

1. Estimate the structural equation by 2SLS to obtain the residuals, $\hat{u}_i$
2. Regress $\hat{u}_i$ on the exogenous variables, the IVs $z_1, \dots, z_k$, and the included covariates $x_q, \dots, x_p$,
3. With the R^2 of the last regression, construct the test statistic nR^2 to evaluate the null hypothesis.

5 Weak Instruments

Let me now say a word about the consequences of choosing an “weak” instrument, that is, a variable that it does not satisfy either conditions (1) or (2). Suppose, for example, that z violates (1), that is, the instrumental variable is not independent of the error term. Considering a model with one (endogenous) regressor,

$$y = \beta x + u$$

it can be shown that,

$$\hat{\beta} \rightarrow \beta + \frac{\text{Cor}(z, u)}{\text{Cor}(z, x)} \frac{\sigma_u}{\sigma_x}$$

where $\text{Cor}(\cdot)$ means correlation.

This is an interesting formula. Even if the correlation between z and u is small, the asymptotic bias can be very large if (a) z , and x are barely correlated, and (b) the noise-to-signal ratio (σ_u/σ_x) is large.

If we compare the asymptotic bias of the poor instrumental variable estimator with the OLS estimator in the case that x is not exogenous,

$$\hat{\beta} \rightarrow \beta + \frac{\text{Cov}(x, u)}{V(x)} = \beta + \frac{\sigma_x \sigma_u \text{Cor}(x, u)}{\sigma_x^2} = \beta + \text{Cor}(x, u) \frac{\sigma_u}{\sigma_x}$$

we may conclude that the OLS estimator is preferred to the 2SLS estimator if

$$\frac{\text{Cor}(z, u)}{\text{Cor}(z, x)} > \text{Cor}(x, u)$$

It is not surprising then that it is often stated that a weak IV can be detected using the first stage regression (e.g., simply looking at the F test or R^2). Standard practice is to check if the instruments are correlated with the endogenous variable by running the endogenous on the IV's and the exogenous variables. An F test on the coefficients on the iv's provides a measure of the information content of the IV's.³

Unfortunately, the problem of weak IV's is typically more complex since it includes other parameters. Consider for instance,

$$\begin{aligned} y_1 &= \beta y_2 + u \\ y_2 &= z' \pi + v, \end{aligned}$$

³A rule of thumb, taken from Stock and Watson (2003), suggests that 10 is the value of the F -statistic that divides weak and valid instruments. For example, if the F statistic exceeds 10, we reject the null hypothesis concluding that the z s are valid instruments for our endogenous variable.

where z is a k -dimensional vector of instruments and $E(uv) = \sigma_{uv}$. The 2SLS estimator is:

$$\hat{\beta} = \frac{y_2' P_z y_1}{y_2' P_z y_2} = \beta + \frac{y_2' P_z u}{y_2' P_z y_2}.$$

Let $q_2 = y_2' P_z y_2 / n$ and note that $y_2' P_z u = \text{tr} y_2' P_z u = \text{tr} P_z y_2' u = \text{tr}(I_k) y_2' u$. Then, one can express the bias of the 2SLS estimator as:

$$\hat{\beta} - \beta \approx \frac{k}{n} \frac{\sigma_{uv}}{q_2} \approx \frac{k}{n R^2} \frac{\sigma_{uv}}{\sigma_{y_2}^2}.$$

This implies that k , the coefficient of determination of the first stage regression R^2 and the correlation between the errors in the equations σ_{uv} play a role in determining the bias of the 2SLS in small samples. Note that as $n \rightarrow \infty$, if k is fixed, we tend to have (asymptotically) zero bias. The literature offers tests and biased-corrected estimators (see e.g., Stock 2002, Hahn and Hausman 2003).

Another problem is related to inference when the correlation between the endogenous variable and the instrument is very small. Again, consider the following model,

$$\begin{aligned} y_1 &= \beta y_2 + u \\ y_2 &= z\pi + v, \end{aligned}$$

where now $k = 1$. In other words, the model is exactly identified and we instrument the endogenous using one instrument. Recall that $\pi = E(z_i y_{2i}) / E(z_i^2)$ and consider that $\pi = \gamma / \sqrt{n}$. In this case,

$$\hat{\beta} - \beta = \frac{\frac{1}{\sqrt{n}} \sum_i z_i u_i}{\frac{1}{\sqrt{n}} \sum_i z_i y_{2i}} \rightsquigarrow \frac{\mathcal{N}(0, V_u)}{Q_z \gamma + \mathcal{N}(0, V_v)},$$

distributed as non-normal for $\gamma \neq 0$ and Cauchy as $\gamma = 0$. Therefore, the IV estimator converges in probability to a non-degenerate random variable. Its distribution is non-normal and consequently inference procedures are inconsistent too (see e.g. Choi and Phillips 1992)

6 IV estimation in non-linear models

Measurement error (ME) in the independent variables causes:

1. bias and inconsistency in parameter estimation for econometric models, and
2. loss of power for detecting stochastic significant relationship between the variables of

interest

The techniques to address the problem of measurement error can be classified in two groups: (a) methods for linear error-in-variables models and (b) methods for non-linear error-in-variables models.

6.1 Classical Errors

This refers to a situation where the true latent variable is uncorrelated with the measurement error. Consider,

$$Y = \alpha + \beta X^* + \epsilon \quad (7)$$

$$X = X^* + u \quad (8)$$

and therefore,

$$Y = \alpha + \beta X + (\epsilon - \beta u) = \alpha + \beta X + v, \quad (9)$$

where $v = \epsilon - \beta u$. Recall that the biased and inconsistent estimator for the slope is,

$$\hat{\beta} = \frac{\sum (X_i - \bar{X}) Y_i}{\sum (X_i - \bar{X})^2} \quad (10)$$

and the intercept is,

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X} \quad (11)$$

Note that,

$$\hat{\alpha} \rightarrow \mathbb{E}(\alpha + \beta X^* + \epsilon) - \frac{\beta \sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_u^2} \mathbb{E}(X^* + u) \quad (12)$$

$$= \alpha + \frac{\beta \sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_u^2} \mathbb{E}(X^*) \quad (13)$$

Remark 1. It is interesting to run a regression for X on Y to find the bounds for the true coefficients α and β . Write,

$$X = -\frac{\alpha}{\beta} + \frac{1}{\beta} Y - \frac{v}{\beta} \quad (14)$$

and therefore,

$$\tilde{b} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (Y_i - \bar{Y})^2}, \quad (15)$$

$$\tilde{\alpha} = \bar{Y} - \tilde{\beta} \bar{X} \quad (16)$$

where $\tilde{\beta} = \tilde{b}^{-1}$. Now we write that

$$\tilde{\alpha} \rightarrow \mathbb{E}(\alpha + \beta X^* + \epsilon) - (\beta + \frac{\sigma_u^2}{\beta \sigma_{X^*}^2}) \mathbb{E}X^* = \alpha - \frac{\sigma_u^2}{\beta \sigma_{X^*}^2} \mathbb{E}X^* \quad (17)$$

Example 1. Note that plim of $\tilde{b} = \text{Cov}(x, y) / \text{Var}(y)$. It is possible to show that

$$\text{plim} \tilde{\beta} = \beta + \sigma_u^2 / \beta \sigma_{X^*}^2$$

As previously discussed, the model with classical errors can be seen as a problem of endogenous regressors. Therefore, IV turns to be a popular approach for estimating such models.

Hausman-type tests can also be used in this type of models. A valid IV could have the form $Z = X^* + v_Z$, with v_Z being an independent error: $\mathbb{E}(Z(\epsilon, u)) = 0$.

Example 2. Consider the previous model, and assume that $\alpha = 0, \beta = 1, \sigma_\epsilon^2 = 0.25, \sigma_u^2 = 1$, and $X^* \in \mathcal{N}(0, 1)$. We can illustrate the attenuation bias using the following two graphs.

Example 3. When we have many columns in X and we observe only one covariate with measurement error, the formula for the attenuation bias is

$$\hat{\beta} \rightarrow \beta \left(1 - \frac{\sigma_u^2}{\sigma_{X^*}^2 (1 - R_{X^*, Z}^2) + \sigma_u^2} \right) \quad (18)$$

where R^2 is the true R^2 .

Example 4. Suppose we observe repeated measurements for individual over time. For instance, assume

$$Y_{it} = \alpha + \beta X_{it}^* + \mu_i + \epsilon_{it} \quad (19)$$

$$X_{it} = X_{it}^* + u_{it} \quad (20)$$

To remove that unobserved heterogeneity, we simply obtain first differences,

$$\Delta Y_{it} = \beta \Delta X_{it}^* + \Delta \epsilon_{it} \quad (21)$$

$$= \beta \Delta X_{it} + \Delta \epsilon_{it} - \beta \Delta u_{it} \quad (22)$$

$$= \beta \Delta X_{it} + \Delta v_{it} \quad (23)$$

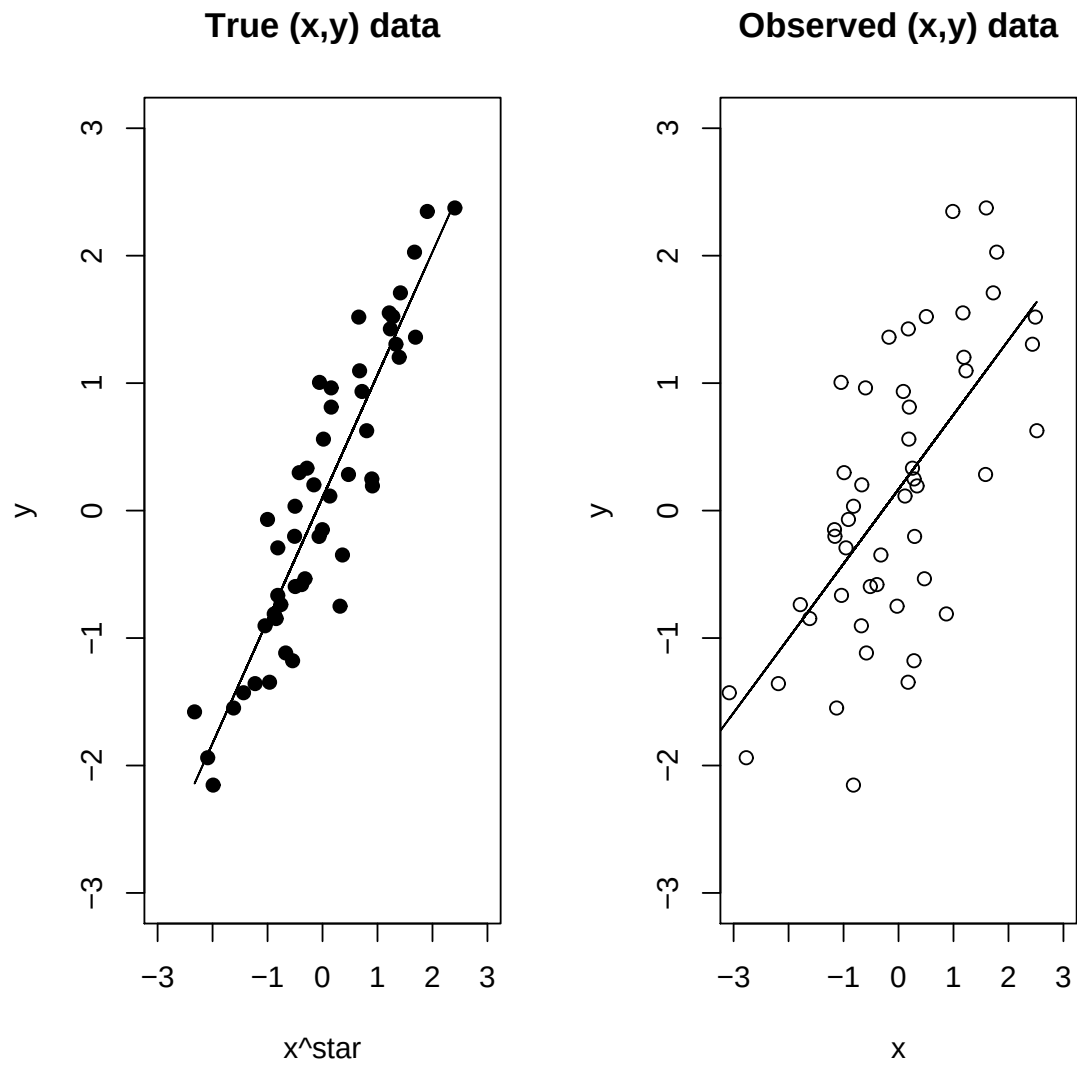


Figure 1: A simple example of the attenuation bias

Therefore,

$$\hat{\beta} = \sum_{i=1}^N \Delta X_{it} \Delta Y_{it} / \sum_{i=1}^N \Delta X_{it}^2 \quad (24)$$

$$= \beta + \frac{1/N \sum_{i=1}^N \Delta X_{it} (\Delta \epsilon_{it} - \beta \Delta u_{it})}{1/N \sum_{i=1}^N \Delta X_{it}^2} \quad (25)$$

Note that,

$$\mathbb{E}(\Delta X_{it})^2 = \mathbb{E}((X_{it}^* + u_{it}) - (X_{it-1}^* + u_{it-1}))^2 \quad (26)$$

$$= \mathbb{E}((X_{it}^* + u_{it})^2 - 2(X_{it}^* + u_{it})(X_{it-1}^* + u_{it-1}) + (X_{it-1}^* + u_{it-1})^2) \quad (27)$$

$$= \sigma_{X^*}^2 + \sigma_u^2 - 2Cov(X_{it}^*, X_{it-1}^*) + \sigma_{X^*}^2 + \sigma_u^2 \quad (28)$$

$$= 2(1 - \rho)\sigma_{X^*}^2 + 2\sigma_u^2 \quad (29)$$

where $\rho = Cor(X_{it}^*, X_{it-1}^*)$. Similarly, it can be shown that

$$\hat{\beta} \rightarrow \beta - \frac{2\beta\sigma_u^2}{2(1 - \rho)\sigma_{X^*}^2 + 2\sigma_u^2} \quad (30)$$

$$= \frac{\beta((1 - \rho)\sigma_{X^*}^2 + \sigma_u^2) - \beta\sigma_u^2}{(1 - \rho)\sigma_{X^*}^2 + \sigma_u^2} \quad (31)$$

$$= \frac{\beta(1 - \rho)\sigma_{X^*}^2}{(1 - \rho)\sigma_{X^*}^2 + \sigma_u^2} \quad (32)$$

6.2 Non-linear Models

It is well known that standard IV assumptions are not enough for identification of nonlinear error-in-variables models. Amemiya (1985) showed that the classical instrumental variable estimator may be inconsistent without additional structure. The literature offers a variety of these restrictions, e.g. from known distributions of measurement errors to stronger forms of IV's.

To illustrate the issues behind classical IV in non-linear settings, let's consider one example introduced in page 912 of Cameron and Trivedi (2005). Consider,

$$Y = \alpha + \beta f(X^*) + \epsilon \quad (33)$$

$$X = X^* + u, \quad (34)$$

where $f(\cdot)$ is a smooth function of a latent variable, X^* . Taylor expansion of $X^* = (X - u)$

around X yields'

$$f(X^*) = f(X - u) = f(X) + \frac{\partial f}{\partial X}(X^* - X) + \frac{1}{2!} \frac{\partial^2 f}{\partial X^2}(X^* - X)^2 \quad (35)$$

$$+ \dots + \frac{1}{n!} \frac{\partial^n f}{\partial X^n}(X^* - X)^n + \dots \quad (36)$$

$$= f(X) - f'u + \frac{1}{2}f''(-u)^2 + \dots + \frac{1}{n!}f^n(-u)^n + \dots \quad (37)$$

Suppose for simplicity that $f = \gamma X + X^2$ is quadratic. In this case, the approximation is exact (e.g., we do not have a reminder term R). Note that $f' = 2X + \gamma$ and $f'' = 2$. Replacing we have that,

$$Y = \alpha + \beta((\gamma X + X^2) - (2X + \gamma)u + \frac{1}{2} \cdot 2u^2) + \epsilon \quad (38)$$

$$= \alpha + \beta\gamma X + \beta X^2 + (\epsilon - 2\beta Xu - \beta\gamma u + \beta u^2). \quad (39)$$

Therefore, IV needs to satisfy:

1. $\mathbb{E}(Z_1 \cdot X) \neq 0$ and $\mathbb{E}(Z_2 \cdot X^2) \neq 0$
2. $\mathbb{E}(Z_j \cdot (\epsilon - 2\beta Xu - \beta\gamma u + \beta u^2)) = 0$,

which is obviously not equivalent to $\mathbb{E}(Z_j \cdot \epsilon) = \mathbb{E}(Z_j \cdot u) = 0$.

GMM, MLE and Tests for Non-linear Restrictions

Lecture 7

This lecture introduces Generalized Method of Moments (GMM), Maximum Likelihood (ML) estimation in a regression setting and discusses tests for non-linear restrictions. As it will be clear later, the Maximum Likelihood method is convenient for dealing with nonlinear models and in some cases non-linear hypothesis. For instance, if you have binary outcome (e.g., $y_i = 1$ if the governor was elected and $y_i = 0$ otherwise) and observable variables (e.g., taxes, state's unemployment rate, etc), it can be shown that the conditional mean model is,

$$\mathbb{E}(y|x) = P(y = 1|x) = F(x'\beta).$$

In the next sections, we discuss the method and associated testing procedures.

1 Generalized Method of Moments (GMM)

Before we present the GMM approach, we briefly review a few simple examples of the Method of Moments. Consider $\{y_i\}$ drawn from an *iid* distribution whose first moment exists (e.g., y is not distributed as Cauchy). In the population,

$$\mathbb{E}(y - \mu) = 0$$

so we can use the corresponding sample moment condition

$$\frac{1}{n} \sum_{i=1}^n (y_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{y}$$

Another example that we already discussed is the IV estimator, that can be defined in terms of the moment condition,

$$\mathbb{E}(z_i u_i) = \mathbb{E}(\mathbb{E}(z_i u_i | z)) = \mathbb{E}(z \mathbb{E}(u_i | z)) = \mathbb{E}(z \times 0) = 0,$$

or alternatively,

$$\mathbb{E}(z_i(y_i - x_i'\beta)) = 0,$$

The IV estimator could be obtained as the solution of the corresponding sample moment condition,

$$\frac{1}{n} \sum_{i=1}^n (z_i(y_i - x_i'\beta)) = 0,$$

yielding the instrumental variable estimator presented in previous lecture,

$$\hat{\beta} = \left(\sum_{i=1}^n z_i x_i' \right)^{-1} \sum_{i=1}^n z_i y_i = (Z'X)^{-1} Z'y$$

We generalize the procedure directly considering models that satisfy the condition,

$$\mathbb{E}(g(w_i, \theta_0)) = 0$$

where $g(\cdot)$ is a function in $\mathbb{R}^l$ and the parameter $\theta_0 \in \Theta \subset \mathbb{R}^p$. For example, $g(w_i, \theta_0)$ had the following linear form

$$z_i(y_i - x_i'\beta),$$

with the minimal requirement of p moment conditions (e.g., $l = p$).

Now suppose than we have l moment conditions with $l > p$. The sample moment counterpart

$$g_n = \frac{1}{n} \sum_{i=1}^n g(w_i, \theta_0)$$

has no solution, but we can select p linear combinations of the sample moments in order to obtain the estimator. Therefore, given $g_n(\theta)$, one can choose a positive semidefinite matrix W_n and define the estimator that minimizes

$$\hat{\theta}_{GMM} = \arg \min_{\theta \in \Theta} \|g_n(\theta)\|_{W_n}^2 = \arg \min_{\theta \in \Theta} g_n(\theta)' W_n g_n(\theta)$$

The matrix $W_n \rightarrow W$ as the sample size goes to infinity with W being positive definite. Of course, the matrix W_n is important for estimation.

1.0.1 Example: Ordinary Least Squares

Consider the case where the errors and the independent variables are independent. For the OLS estimator the conditions are,

$$\mathbb{E}(x_i(y_i - x_i'\beta)) = 0$$

The sample moments conditions are,

$$g_n(\beta) = \frac{1}{n} \sum_{i=1}^n x_i(y_i - x_i'\beta)$$

The number of moment conditions is equal to the number of unknown parameters. In this case, we say that the model is just or exactly identified. The IV case, illustrated above, also represents a case where a model is exactly identified.

1.0.2 Example: Two Stage Least Squares

For the 2SLS, we may have more moment conditions than parameters (e.g., $\dim(z_i) > \dim(x_i)$), but a proper matrix W will be enough to obtain the estimator,

$$\arg \min g_n(\beta)' W_n g_n(\beta)$$

with $W_n = (Z'Z)^{-1}$ and $g_n = Z'(y - X\beta)$. We note that

$$X'ZWZ(y - X\beta) = 0,$$

implies

$$\hat{\beta} = (X'ZWZ'X)^{-1}X'ZWZ'Xy = (X'P_zX)^{-1}X'P_zy$$

The solution depends on the matrix W , and of course, the solution of the minimization problem is the same for W 's that differ by a constant of proportionality. Note that this is related with the assumption of homocedastic errors. What should be the weighting matrix under heterocedasticity? Let S be the covariance matrix of the moment condition g ,

$$S = \frac{1}{n} \mathbb{E}(Z'uu'Z) = \frac{1}{n} Z'\Omega Z$$

To obtain the optimal GMM estimator in this case, we need:

1. Estimate $\hat{\beta}$ by IV methods to obtain $\hat{u}$.
2. Use $\hat{u}$ to find,

$$\hat{W} = \hat{S}^{-1} = \frac{1}{n} (Z'\hat{\Omega}Z)^{-1}$$

where $\hat{\Omega} = \text{diag}(\hat{u}_i^2)$.

3. Find the feasible, optimal GMM as

$$\hat{\beta} = (X'Z(Z'\hat{\Omega}Z)^{-1}Z'X)^{-1}X'Z(Z'\hat{\Omega}Z)^{-1}Z'Xy$$

2 Further Considerations

If the model is identified, the estimator is consistent and asymptotically normally distributed under regularity conditions,

1. The first regularity condition assumes that

$$\frac{\partial g_n(\theta)}{\partial \theta'} \rightarrow \frac{\partial \mathbb{E}(g(\theta))}{\partial \theta'} = G$$

2. Also we assume the standard square-root consistency,

$$\sqrt{n}g_n(\theta) \rightsquigarrow \mathcal{N}(0, S^{-1})$$

Then, the estimator is asymptotically normally distributed,

$$\sqrt{n}(\hat{\theta} - \theta) \rightsquigarrow \mathcal{N}(0, (G'WG)^{-1}G'WSW'G(G'WG)^{-1})$$

If we choose W_n such that $W_n \rightarrow S^{-1} = W$, we have that the asymptotic variance is simply,

$$(G'S^{-1}G)^{-1},$$

the case that it is known as efficient GMM (Hansen 1982) when S is known. This is perhaps a disadvantage of the estimator since requires to know the asymptotic covariance matrix. The solution is to estimate it, but this idea may fail in small and intermediate samples. If the sample size is not large enough, or if the moment conditions is large, the estimation of the asymptotic covariance matrix in the first step can result in bad estimates in the second step. The problem has been documented in the literature, and various solutions (e.g. simultaneous estimation) has been proposed.

3 ML Estimation

Let $\{(y_i, x_i) : i = 1 \dots n\}$ be an independently and identically distributed (*i.i.d.*) sample of the response y and the independent variables x .

We can define the likelihood function as

$$\mathcal{L}(\theta) = f(y, x|\theta),$$

but since our goal is to model the behavior of y in terms of x , we consider directly the conditional density of y_i ,

$$f(y_i|x_i, \theta_0)$$

Then we can define the conditional log-likelihood function as

$$\ell(\theta; y_i | x_i) \equiv \log f(y_i | x_i, \theta)$$

Considering an iid sample of n observations $\{(y_i, x_i)\}$, the joint conditional density function is basically

$$\prod_{i=1}^n f(y_i | x_i, \theta) = f(y_1 | x_1, \theta) \dots f(y_n | x_n, \theta)$$

and consequently the log-likelihood function is

$$\ell(\theta; y | x) = \log\left(\prod_{i=1}^n f(y_i | x_i, \theta)\right) \equiv \sum_{i=1}^n \log f(y_i | x_i, \theta)$$

Example 1. (Normal Location Scale) Suppose that u_i is distributed as $\mathcal{N}(\beta_0, \sigma^2)$. Then the probability density function is

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(u_i - \beta_0)^2}{2\sigma^2}}$$

If the observations are independent, we have

$$(2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum_i (u_i - \beta_0)^2},$$

and consequently the log-likelihood as,

$$\ell(\beta_0, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (u_i - \beta_0)^2$$

Example 2. (The regression model) Consider the classical regression model,

$$y_i = x_i' \beta + u_i$$

where y and x are iid, jointly distributed random variables. Transforming variables will get the log-likelihood as

$$\sum_{i=1}^n \log f(y_i | x_i, \theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i' \beta)^2$$

where $\theta = (\beta, \sigma^2)$.

Example 3. (Poisson density) Consider now that the dependent variable y_i takes non-negative discrete value (e.g. number of visits to the doctor). The Poisson density function

may be a natural direction,

$$f(y|\lambda) = \frac{e^{-\lambda}\lambda^y}{y!}; \quad y = 0, 1, 2, \dots$$

The usual specification in regression analysis is $\lambda = \exp(x'\beta)$, leading to a log-likelihood function of the form

$$\sum_{i=1}^n \log f(y_i|x_i, \theta) = - \sum_{i=1}^n \exp(x_i'\beta) + \sum_{i=1}^n y_i x_i'\beta - \sum_{i=1}^n \log y_i!$$

3.1 Maximum Likelihood Estimator

Consider the following assumptions:

1. The model $y_i = m(x_i, u_i, \theta)$ is correctly specified where $m(\cdot)$ is a known function that can be linear (e.g., $y = x'\beta + u$).
2. $\{(y_i, x_i) : i = 1 \dots n\}$ is an i.i.d. sample
3. Θ is a compact subspace in $\mathbb{R}^p$ that contains θ_0
4. The density function $f(y_1|x, \theta)$ is continuous and has continuous second order derivatives over a compact set Θ .
5. The likelihood function has a unique solution.

The MLE principle is to choose an estimator of a parameter of the conditional distribution of y , θ_0 , that maximizes the probability of observing the actual sample. The maximum likelihood estimator (MLE) is the estimator that maximizes the (log)likelihood function,

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \ell(\theta; y|x)$$

where Θ is a compact subspace in $\mathbb{R}^p$ that contains θ .

Example 4. (Linear Regression) Consider the log-likelihood function we obtained before using matrix notation,

$$\text{const} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta)$$

Taking the first order condition,

$$\begin{aligned}\frac{\partial \ell}{\partial \beta} &= \frac{1}{\sigma^2} X'(y - X\beta) = 0 \\ \frac{\partial \ell}{\partial \sigma^2} &= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} (y - X\beta)'(y - X\beta) = 0\end{aligned}$$

Solving the system of equations, we find that the MLE estimators are

$$\hat{\beta} = (X'X)^{-1}X'y \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n}(y - X\hat{\beta})'(y - X\hat{\beta})$$

We know that the estimator $\hat{\beta}$ is unbiased because it is identical to the OLS estimator. In contrast, the estimator of the variance parameter $\hat{\sigma}^2$ is biased. (Why?). Also,

$$\begin{aligned}\frac{\partial^2 \ell}{\partial \beta \partial \beta'} &= -\frac{X'X}{\sigma^2} \\ \frac{\partial^2 \ell}{\partial \beta \partial \sigma^2} &= -\frac{X'(y - X\beta)}{\sigma^4} \\ \frac{\partial^2 \ell}{\partial \sigma^2 \partial \beta'} &= -\frac{(y - X\beta)'X}{\sigma^4} \\ \frac{\partial^2 \ell}{\partial \sigma^2 \partial \sigma^2} &= \frac{n}{2\sigma^4} - \frac{(y - X\beta)'(y - X\beta)}{\sigma^6}\end{aligned}$$

The expectation of the $p \times p$ matrix of second derivatives is called the information matrix,

$$\begin{aligned}I(\theta_0|X) &= -\mathbb{E}(\partial^2 \ell / \partial \theta \partial \theta' | X) \\ &= -\begin{bmatrix} \mathbb{E}(\partial^2 \ell / \partial \beta \partial \beta') & \mathbb{E}(\partial^2 \ell / \partial \beta \partial \sigma^2) \\ \mathbb{E}(\partial^2 \ell / \partial \sigma^2 \partial \beta') & \mathbb{E}(\partial^2 \ell / \partial (\sigma^2)^2) \end{bmatrix} \\ &= \begin{bmatrix} \frac{X'X}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}\end{aligned}$$

Note that the off diagonal elements of the information matrix $I(\theta)$ are zero suggesting that $\hat{\beta}$ and $\hat{\sigma}^2$ are “independent”. The matrix of second derivatives is n.s.d.

Example 5. (Poisson density) The Poisson case has no explicit solution, thus numerical methods are useful in this case. By differentiating the likelihood function, we obtain $\hat{\beta}$ as the solution of

$$\sum_{i=1}^n y_i x_i - \sum_{i=1}^n \exp(x_i \hat{\beta}) x_i = 0$$

The previous case it is common in empirical analysis. Sometimes obtaining the MLE estimator gets complicated not yielding analytic solutions. Below I will briefly discuss alternatives

to overcome the problem.

3.2 Efficiency

Let define the score function as,

$$S_i = \frac{\partial \ell_i(\theta)}{\partial \theta} = \left(\left(\frac{\partial \ell_i(\theta)}{\partial \theta_1} \right), \dots, \left(\frac{\partial \ell_i(\theta)}{\partial \theta_p} \right) \right)',$$

with $\mathbb{E}(S_i(\theta_0)) = 0$, and the Hessian matrix,

$$H_i = \frac{\partial^2 \ell_i(\theta)}{\partial \theta \partial \theta'} = \frac{\partial S_i}{\partial \theta},$$

a symmetric $p \times p$ matrix of second derivatives. Evaluated at the true parameter θ_0 , we have that

$$-\mathbb{E}[H(\theta)] = -\mathbb{E} \left[\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta'} \right] = \mathbb{E} \left[\left(\frac{\partial \ell(\theta)}{\partial \theta} \right) \left(\frac{\partial \ell(\theta)}{\partial \theta'} \right)' \right] = I(\theta_0),$$

This property is important for empirical analysis. It is time consuming to compute the second derivative of a likelihood function, but the previous expression suggest an alternative, perhaps easier way: compute the product of the two score functions. We need to be cautious, though. The previous analysis holds when we know the conditional distribution of y .

In the case of iid samples,

$$-\sum_{i=1}^n \mathbb{E} \left(\frac{\partial^2 \ell_i(\theta)}{\partial \theta \partial \theta'} \right) = -n \mathbb{E} \left(\frac{\partial^2 \ell_i(\theta)}{\partial \theta \partial \theta'} \right) = n \mathbb{E} \left[\left(\frac{\partial \ell_i(\theta)}{\partial \theta} \right) \left(\frac{\partial \ell_i(\theta)}{\partial \theta'} \right)' \right] = n \Omega(\theta),$$

indicating that the information matrix is n times the information contained in one observation. Note that Ω is the variance of the score function.

It can be shown that the MLE estimator is consistent meaning that

$$\hat{\theta} \rightarrow \theta,$$

and asymptotically normal

$$\sqrt{n}(\hat{\theta} - \theta_0) \rightsquigarrow \mathcal{N}(0, \Omega(\theta)^{-1})$$

If the number of observations is sufficiently large, the estimator has asymptotic variance that can be approximated by the inverse of the information contained in one observation. This results shows the efficiency of the MLE estimator, which reaches the Cramer-Rao lower

bound (the lower bound of the variance of unbiased estimators),

$$V(\tilde{\theta}) \geq (nI(\theta))^{-1},$$

for any unbiased $\tilde{\theta}$ estimator of θ_0 , provided that the information matrix is non-singular. Note that MLE is attractive for estimation of parametric models since it achieves the lower bound.

4 Testing procedures

This section considers tests of linear and non-linear hypothesis. We review three tests based on the likelihood principle and we provide the basis to consider tests when classical methods fails. Suppose we have an iid sample of n observations $\{(y_i, x_i); i = 1, \dots, n\}$. We consider test of hypothesis of the form,

$$h(\theta) = 0$$

where $h(\cdot)$ is a $r \times 1$ vector function of θ with $r \leq p$. We assume that $H(\theta) = \partial h(\theta)/\partial \theta$, a $p \times r$ matrix, is well defined and it has full column rank

$$\text{rank}(H(\theta)) = r,$$

which means that the restrictions imposed are independent. Before considering the test, let me define a couple of important concepts. We denote the restricted parameter space

$$\Theta_0 = \{\theta \in \Theta | h(\theta) = 0\},$$

implying that $\Theta_0 \subset \Theta$. Recall that the MLE estimator is defined as,

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \{\ell(\theta)\}$$

Also, we can define the restricted MLE estimator as the argument that maximizes the restricted log-likelihood function,

$$\tilde{\theta} = \arg \max_{\theta \in \Theta_0} \ell(\theta)$$

or equivalently,

$$\max_{\theta \in \Theta_0} \ell(\theta) \quad \text{subject to} \quad h(\theta) = 0$$

4.1 Likelihood Ratio Test

The test is defined as,

$$\text{LR} = 2\{\ell(\hat{\theta}) - \ell(\tilde{\theta})\}$$

The intuition behind the test is that if the null hypothesis is true, the likelihood functions should be similar. Therefore, higher values of LR implies higher chances of being rejecting the null hypothesis H_0 .

4.2 Lagrange Multiplier Test

$$LM = \frac{\partial \ell(\tilde{\theta})}{\partial \theta'} I(\tilde{\theta})^{-1} \frac{\partial \ell(\tilde{\theta})}{\partial \theta},$$

which is motivated by the fact that the score function $\partial \ell(\hat{\theta}/\partial \theta) = 0$. Therefore, if the null hypothesis H_0 is true, we expect that the maximum should occur at $\partial \ell(\tilde{\theta}/\partial \theta) \approx 0$ failing to reject the null.

4.3 Wald Test

The test is defined as,

$$W = h(\hat{\theta})' \left[H(\hat{\theta})' I(\hat{\theta})^{-1} H(\hat{\theta}) \right]^{-1} h(\hat{\theta})$$

If H_0 is true, $h(\hat{\theta})$ is close to zero. The test statistic is asymptotically χ_r^2 where r denotes the rank of the matrix H (i.e. $r = \text{rank}(H)$ -the number of restrictions imposed by H_0 -).

Example 6. (Tests of Nonlinear Restrictions) Consider testing

$$H_0 : h(\theta) = \frac{\theta_1}{\theta_2} - c = 0$$

Then the Wald statistics is

$$W = \left(\frac{\theta_1}{\theta_2} - c \right) \left[\left(\frac{1}{\theta_2} - \frac{\hat{\theta}_1}{\hat{\theta}_2^2} \quad 0 \right) \begin{bmatrix} \hat{v}_{11} & \hat{v}_{12} & \dots & \dots \\ \cdot & \hat{v}_{22} & \dots & \dots \\ \dots & \dots & \dots & \dots \end{bmatrix} \begin{pmatrix} \frac{1}{\hat{\theta}_2} \\ -\frac{\hat{\theta}_1}{\hat{\theta}_2^2} \\ 0 \end{pmatrix} \right]^{-1} \left(\frac{\theta_1}{\theta_2} - c \right)$$

where 0 is a $(r-2) \times r$ matrix of zeros and v_{jj} denote the jj -th element of the estimated asymptotic covariance matrix. Under the null W is asymptotically distributed as χ_1^2 .

The problem with this test is that it depends on the estimated asymptotic covariance matrix. There are alternatives for non-linear restrictions to be discussed in the next courses (e.g., Delta Method, Bootstrap).

Example 7. (Restricted MLE) Consider now a log-likelihood of the form,

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta) + 2\lambda'(R\beta - \delta),$$

where λ is a potentially $p \times 1$ vector. Assuming we know the value of σ^2 , the solution can be written as,

$$\frac{\partial \ell}{\partial \beta} = 0 \Rightarrow -2X'y + 2X'X\tilde{\beta} + 2R'\lambda = 0$$

$$\frac{\partial \ell}{\partial \lambda} = 0 \Rightarrow R\tilde{\beta} = \delta$$

In other words,

$$\begin{bmatrix} X'X & R' \\ R & 0 \end{bmatrix} \begin{bmatrix} \tilde{\beta} \\ \lambda \end{bmatrix} = \begin{bmatrix} X'y \\ \delta \end{bmatrix}$$

Therefore,

$$(\tilde{\beta}', \lambda)' = \begin{bmatrix} X'X & R' \\ R & 0 \end{bmatrix}^{-1} \begin{pmatrix} X'y \\ \delta \end{pmatrix}$$

Applying inverse-partitioned matrix formulae, we obtain,

$$\begin{aligned} \tilde{\beta} &= (X'X)^{-1}X'y - (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1} \\ &\quad R(X'X)^{-1}X'y + (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}\delta \\ &= \hat{\beta} - (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}R\hat{\beta} + (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}\delta \\ &= \hat{\beta} - D(R\hat{\beta} - \delta) \end{aligned}$$

where $D = (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}$. Note that

$$E(\tilde{\beta}) = E(\hat{\beta} - D(R\hat{\beta} - \delta)) = \beta - D(R\beta - \delta) = \beta,$$

if $R\beta - \delta = 0$ (the null hypothesis is true). If it is false, $\tilde{\beta}$ is biased.

In terms of the variance of $\tilde{\beta}$, we have that,

$$\begin{aligned} V(\tilde{\beta}) &= V(\hat{\beta} - D(R\hat{\beta} - \delta)) = V(\hat{\beta} - DR\hat{\beta}) = V((I - DR)\hat{\beta}) \\ &= (I - DR)V(\hat{\beta})(I - DR)' \\ &= \sigma^2(I - DR)(X'X)^{-1}(I - DR)' \\ &= \sigma^2(X'X)^{-1} - \sigma^2[DR(X'X)^{-1} + (X'X)^{-1}R'D' - DR(X'X)^{-1}R'D'] \end{aligned}$$

Note that,

$$\begin{aligned} DR(X'X)^{-1} &= (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}R(X'X)^{-1} \\ &= (X'X)^{-1}R'D'. \end{aligned}$$

and

$$\begin{aligned} DR(X'X)^{-1}R'D' &= (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}(R(X'X)^{-1}R')D' \\ &= (X'X)^{-1}R'D' \end{aligned}$$

Therefore,

$$\begin{aligned} V(\tilde{\beta}) &= \sigma^2(X'X)^{-1} - \sigma^2 DR(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1}(I - DR) \leq V(\hat{\beta}). \end{aligned}$$

The solution for the lagrange multiplier is,

$$\begin{aligned} \tilde{\lambda} &= (R(X'X)^{-1}R')^{-1}R(X'X)^{-1}X'y - (R(X'X)^{-1}R')^{-1}\delta \\ &= W(R\hat{\beta} - \delta). \end{aligned}$$

It is easy to see that $E(\tilde{\lambda}) = 0$ under the null and $V(\tilde{\lambda}) = \sigma^2 W$.

5 Model Selection

In some situations, typically time series, we may not know the dimension of the model (e.g., how many parameters our linear model should have). This important aspect of modeling in statistics and econometrics was early addressed by Akaike (1969) and Schwartz (1978).

In a regression setting, the likelihood can be written as

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)$$

By plugging in $\hat{\beta}$ and $\hat{\sigma}^2 = \mathcal{S}/n$, we obtain the concentrated likelihood function

$$\ell(\hat{\sigma}^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2} - \frac{n}{2}\log\hat{\sigma}^2$$

suggesting that for a given sample size, the likelihood can be increased by adding covariates to the model to reduce the residual sum of squares $\mathcal{S}$.

Akaike (1969) proposed to choose a model from a set that performs well on the basis of forecasting. His proposal is based on the following function,

$$\text{AIC}(j) = \ell_j(\hat{\theta}) - p_j$$

where $\ell_j(\hat{\theta})$ is the likelihood corresponding to the j^{th} model maximized over $\theta \in \Theta_j$ (typically we have $p_1 < p_2 < \dots < p_n$). AIC stands for Akaike Information Criteria. The basic idea of

the additional term in AIC, called a penalty term, is to bring a rule over the fit of various specifications by “penalizing” an increase in the number of regressions. Akaike’s model selection rule was to evaluate $AIC(j)$ over j models, choosing the model j^* that maximize $AIC(j)$.

A latter criterion, which is known as Schwarz information criterion, is based on a Bayesian approach to the problem analyzed in Akaike (1969). Schwarz proposed

$$SIC(j) = \ell_j(\hat{\theta}) - \frac{1}{2}p_j \log n$$

Note that maximizing SIC

$$\ell_j - \frac{1}{2}p_j \log n$$

is equivalent to minimizing

$$\log \hat{\sigma}_j^2 + \frac{p_j}{n} \log n$$

It is possible to show that the test statistics,

$$T_n = 2(\ell_j(\hat{\theta}_j) - \ell_i(\hat{\theta}_i)) \rightsquigarrow \chi_{p_j - p_i}^2$$

for $p_j > p_i = p^*$. (Note that model i is the true model). Classical hypothesis testing would suggest to reject the null that the smaller model i is the true model if and only if the value of T_n exceeds a $\chi_{p_j - p_i}^2$ critical value. On the other hand, SIC may choose j over i if

$$2 \frac{(\ell_j - \ell_i)}{p_j - p_i} > \log n, \tag{1}$$

because

$$\begin{aligned} 0 &= (\ell_j(\hat{\theta}) - \frac{p_j}{2} \log n) - (\ell_i(\hat{\theta}) - \frac{p_i}{2} \log n) \\ &= (\ell_j(\hat{\theta}) - \ell_i(\hat{\theta})) - \frac{1}{2}(p_j - p_i) \log n. \end{aligned}$$

Note that the number of observations is acting like the critical value of the test. This is closely connected with the classical F test, since it is possible to show that, considering for

simplicity $p_j - p_i = 1$,

$$\begin{aligned}
 (\ell_i - \ell_j) &= -n(\log \hat{\sigma}_i^2 - \log \hat{\sigma}_j^2) \\
 &= n(\log \hat{\sigma}_j^2 - \log \hat{\sigma}_i^2) \\
 &= n \log(\hat{\sigma}_j^2 / \hat{\sigma}_i^2) \\
 &= n \log(1 + \hat{\sigma}_j^2 / \hat{\sigma}_i^2 - 1) \\
 &= n \log(1 + a)
 \end{aligned}$$

where $a = (\hat{\sigma}_j^2 / \hat{\sigma}_i^2 - 1)$. Using $\log(1 + a) \approx a$ for small a , we have

$$2(\ell_i - \ell_j) \approx \frac{n(\hat{\sigma}_j^2 - \hat{\sigma}_i^2)}{\hat{\sigma}_i^2} > \log n \quad (2)$$

The LHS of equation (2) can be interpreted as an F test since it can be written approximately as

$$\frac{\text{SSR}_r - \text{SSR}_u}{\text{SSR}_u}.$$

We will use AIC and BIC as model selection devices for time series models. You can find a preliminary example in the tutorial to this lecture notes.

6 A (final) digression

The previous framework connects nicely with Big Data analytic. Consider the following estimator,

$$\hat{\beta} = \arg \min \left[\sum_{i=1}^n (y_i - x_i' \beta)^2 + P(\beta, \lambda) \right] \quad (3)$$

where $P(\beta, \lambda)$ is a penalty function and λ is a tuning parameter. For example, the penalty function can be

$$P(\beta, \lambda) = \lambda \sum_{j=1}^p |\beta_j|^q \quad (4)$$

for $q = \{1, 2\}$. The model here is sparse in the coefficients β 's and the solution of the optimization problem depends on the value of the shrinkage parameter and the penalty function.

For easily differentiable penalty functions, for instance we have that

$$\begin{aligned}
 u'u + \lambda\beta'\beta &= (y - X\beta)'(y - X\beta) + \lambda\beta'\beta \\
 &= (y' - \beta'X')(y - X\beta) + \lambda\beta'\beta \\
 &= y'y - 2\beta'X'y + \beta'X'X\beta + \lambda\beta'\beta
 \end{aligned}$$

Assuming $\lambda/2$ given, the FOC $-X'y + X'X\beta + \lambda\beta = 0$. Therefore,

$$\hat{\beta} = (X'X + \lambda I)^{-1}X'y.$$

It can be shown that,

$$\begin{aligned}
 E(\hat{\beta}) &= (X'X + \lambda I)^{-1}\beta \\
 V(\hat{\beta}) &= (X'X + \lambda I)^{-1}\sigma^2(X'X)(X'X + \lambda I)^{-1}.
 \end{aligned}$$

A Appendix:

Recall that $1 = \int f(x)dx$. Differentiating both sides,

$$\begin{aligned}
 0 &= \int \frac{1}{f} f_{\theta}(x) f dx \\
 0 &= \int \frac{\partial \log f}{\partial \theta} f dx
 \end{aligned}$$

We know that

$$\begin{aligned}
 0 &= \int \frac{\partial \log f}{\partial \theta} f dx \\
 0 &= \int \left[\frac{\partial^2 \log f}{\partial \theta \partial \theta'} f + \frac{\partial \log f}{\partial \theta} \frac{\partial f}{\partial \theta} \right] dx \\
 0 &= \int \left[\frac{\partial^2 \log f}{\partial \theta \partial \theta'} f + \left(\frac{\partial \log f}{\partial \theta} \right) \left(\frac{\partial \log f}{\partial \theta'} \right) f \right] dx \\
 0 &= \int \left[\frac{\partial^2 \log f}{\partial \theta \partial \theta'} \right] f dx + \int \left[\left(\frac{\partial \log f}{\partial \theta} \right) \left(\frac{\partial \log f}{\partial \theta'} \right) \right] f dx
 \end{aligned}$$